All 38 matched cases

Reported coding: first AI-assisted coding pass. Second-pass judgments remain in the companion CSV. Source categories are descriptive context, not the Work Left code.

Each case shows the nearby conversation followed by three actual tutor responses, their source, and the reported codes. W describes mathematical work left for the learner: W0 none or repetition, W1 calculation, W2 connection or explanation, W3 strategy or argument. R describes information supplied: R0 none, R1 general guidance, R2 a task-specific step, R3 the complete answer. NA means unclear. The separate coding manual gives the complete rules. The full earlier conversation is preserved in the companion JSON file.

E01

Recent conversation

tutor (T73): Can you tell me what that star or what the number that would go in the star is?

tutor (T74): [PAUSE: 16 seconds]

student (T74): Twelve.

tutor (T75): Twelve? Okay. Um, why do you think it's twelve?

student (T76): Because

tutor (T77): [PAUSE: 25 seconds] Student is thinking.

tutor (T77): Do you know why, Diego?

student (T78): No, I'm trying to figure it out.

tutor (T79): Okay, okay.

tutor (T80): Just remember this. Um, in multiplication, if it's only multiplication, right? The order of the numbers don't matter, okay? So keep that in mind um when you're doing this. That might help you a little bit.

tutor (T81): [PAUSE: 7 seconds]

tutor (T81): Or think about the parentheses, what's in there. Because you got to make the two sides like be the same, you know? So figure out or think about what the parentheses is doing there.

student (T82): It's sectioning off parts of the equation.

tutor (T83): Yeah. So, you see that the twenty-five is on both side of the equations, right?

student (T84): Yes.

tutor (T85): So, to do fourteen, right? You need basically this side right here needs to be twenty-five times fourteen.

student (T86): So the answer's fourteen.

tutor (T87): Well, good, it's you're close, right? You're really close. I can see how it could be fourteen, but um it's not the right answer because there's that two over there, right?

student (T88): Oh yeah.

tutor (T89): And you you got to do what's in the parentheses first. So to get twenty-five times fourteen on this side, you need twenty-five and then a number that's gonna come from what's in the parentheses.

tutor (T90): And that number that's in the parentheses needs to be a fourteen.

tutor (T91): [PAUSE: 13 seconds]

tutor (T91): [PROBLEM_CHANGE: p1 - '25 x 14 = 25 x (2 x ★)' (start)] The problem on the screen is "25 x 14 = 25 x (2 x ★)". The question is "What should replace the star to make this calculation correct?". The options are A) 7, B) 14, C) 21, D) 28.

student (T91): I don't really get it.

GPT-5.5: general tutoring instructions | E01-A | W0 / R3

No worries — let’s slow it way down.

You were actually really close when you said 14, because we do need to make a 14.

Look at it like this:

25 × 14 = 25 × (2 × ★)

Both sides already have **25 ×**.

So the part that comes after the 25 needs to match:

14 = 2 × ★

So now the question is:

2 times what number equals 14?

2 × 7 = 14

So the star should be **7**.

Let’s check it:

25 × (2 × 7)

2 × 7 = 14

So it becomes:

25 × 14

Same as the left side. So the answer is **7**. Does that part make more sense now?

Evidence: So the star should be **7**.

Rationale: The response answers its own two-times question and completes the check; only understanding assent remains.

Human tutor | E01-B | W1 / R2

You don't get it? Okay.

So, forget this one side of the equation. Forget this, right? Um, you want to make, um, the part where the star is, you want to make that entire side, um, be 25 times 14.

And you remember how last time we talked about how whatever is in the parentheses comes first, right? No matter what, that we always do what's in there first.

So we already have the 25 on both sides. So we don't need to think about that anymore.

What we want is to have what's in the parentheses be 14, right?

Or, and we already have a two over there. So we need two times a random number, that's the, the ones we're going to choose from, and those two numbers have to multiply to be 14.

Evidence: those two numbers have to multiply to be 14

Rationale: The learner still finds the factor paired with 2; the target product and operation are supplied.

GPT-5.5: explicit scaffolding and rigor instructions | E01-C | W1 / R2

No worries — let’s slow it way down.

The left side is:

25 × 14

The right side is:

25 × (2 × ★)

Since both sides already have **25 ×**, the part in the parentheses needs to become **14**.

So we can turn it into a smaller question:

2 × ★ = 14

What number times 2 makes 14?

Evidence: What number times 2 makes 14?

Rationale: The reduced equation is supplied but its missing factor still requires calculation.

E02

Recent conversation

tutor (T12): Well, we can get started right away. Um, so, you're going to do, remember we're going to be practicing on your quick, uh, what's it called? Your multiplication, division, just like solving them quick at the end.

tutor (T13): Um, first, I'm going to show you 15 problems. They're pretty easy, okay? Like, they're, you should get most of them done. And what I'm going to do is I'm going to put on a 20-minute timer and see how many you can get done.

tutor (T14): The minimum you should get done is 10 questions, okay?

student (T15): Okay.

tutor (T16): So as soon as, um, you see them on your screen, get started right away, okay? And remember, you can always ask questions too.

student (T17): Okay.

tutor (T18): All right, here they are.

tutor (T19): [SCREEN_UPDATE] The tutor's video feed is replaced by a black screen.

tutor (T19): I'll check back in at 3:25.

tutor (T20): [PAUSE: 17 seconds]

tutor (T20): [PROBLEM_CHANGE: p1-15 - Math Problems (start)] A list of 15 math problems appears on the left side of the screen. The problems are:

1. What is 6 x 4?

2. Multiply: 12 x 3

3. Add: 7 + 82

4. What is the value of x if x + 8 = 12?

5. What is the perimeter of a rectangle with length 8 cm and width 3 cm?

6. Add: 1/2 + 1/2

7. What is 9 x 2?

8. Divide: 24 / 6

9. What is 3/3 equal to?

10. What is the value of x if 2x = 10?

11. Subtract: 15 - 9

12. What is the perimeter of a square with sides of 6 inches?

13. Multiply: -8 x 7

14. What is the value of x if x - 3 = 4?

15. Add: 1M + 1M

Tutor's cursor moves to the right of the problem list, and a name tag "Kayla" appears.

tutor (T20): You have about 10, 12 minutes left, all right? Just a reminder.

student (T21): Okay.

tutor (T22): [PAUSE: 3 seconds]

tutor (T22): [SCREEN_INTERACTION: tutor - moves cursor] Tutor's cursor moves away, and the "Kayla" name tag disappears.

tutor (T22): [PAUSE: 356 seconds]

tutor (T22): Hi Kayla. You still have some time, but I was wondering which question are you on?

student (T23): I'm on eight.

tutor (T24): [SCREEN_INTERACTION: tutor - pointing] The tutor's cursor moves to point at question 8.

tutor (T24): All right. All right. Remember, try to go as fast as you can. Um, you should, I think you should be a little farther than eight, but just keep going. Also for number 10, um, go ahead and skip number 10, okay?

student (T25): Okay.

tutor (T26): Just keep going with the rest. Keep going, real fast.

tutor (T27): [PAUSE: 3 seconds]

student (T27): I need help on nine.

GPT-5.5: explicit scaffolding and rigor instructions | E02-A | W1 / R2

Sure. For #9, think of it as **3 divided by 3**.

If you have **3 things** and you put them into groups of **3**, how many groups do you have?

Evidence: how many groups do you have?

Rationale: The learner executes the specified grouping calculation 3 divided by 3.

GPT-5.5: general tutoring instructions | E02-B | W0 / R3

Number 9 means 3 divided by 3. If you split 3 into 3 equal groups, each group has 1.

So 3/3 = 1. Write 1 and keep going.

Evidence: So 3/3 = 1. Write 1 and keep going.

Rationale: Only copying the supplied quotient is specifically elicited; no particular later problem is posed.

Human tutor | E02-C | W0 / R0

Um, you can go ahead and skip nine for now too.

Evidence: skip nine for now too

Rationale: Skipping removes the current demand without posing a new mathematical action.

E03

Recent conversation

student (T177): B?

tutor (T178): Mm-mm.

tutor (T179): [PAUSE: 1 second]

student (T179): D?

tutor (T180): Okay.

student (T181): D?

tutor (T182): Did you say B or D?

student (T183): D.

tutor (T184): Yeah.

tutor (T185): [PAUSE: 2 seconds] Tutor writes "34 - ? = 15" on the whiteboard.

Tutor writes "35 - ? = 16" and "36 - ? = 17" on the whiteboard.

tutor (T185): What they're asking you to do is, um, this, you do the same thing to all the numbers. So, that means that thirty-four minus nineteen... um, hold on, actually. Let me do it the way they're showing it to you.

tutor (T186): So, thirty-four minus this question mark over here is equal to fifteen.

tutor (T187): And you know it's minus because if your input, like if the number you start with is bigger than the number you end with, then you subtracted, right? And if the number you start with is smaller than the one you end with, you add it. Does that make sense?

student (T188): Yeah.

tutor (T189): Yeah. So we know it's a subtract, so already we can like eliminate C. And then thirty-five minus this question mark over here is equal to sixteen. And then thirty-six minus this question mark right here is equal to seventeen.

tutor (T190): Now, because all these numbers are the same, right? This question mark is going to be the same every time. They're all the same number. That means that if you want to find out what it is, you can pretty easily just do thirty-four minus fifteen, thirty-five minus sixteen, or thirty-six minus seventeen, whichever you want to do. So, if you were to do thirty-six minus seventeen, what would it give you?

tutor (T191): [PAUSE: 3 seconds]

student (T191): Huh?

tutor (T192): Thirty-six minus seventeen. What would it give you?

tutor (T193): [PAUSE: 5 seconds]

student (T193): Can you repeat that one more time?

tutor (T194): Oh, I was saying that thirty-four minus the same number, like these question marks on the screen right here are all going to be the same number. So you can find this by doing thirty-four minus fifteen, thirty-five minus sixteen, or thirty-six minus seventeen. So if you did thirty-six minus seventeen, what would you get?

tutor (T195): [PAUSE: 11 seconds]

student (T195): Nineteen.

GPT-5.5: general tutoring instructions | E03-A | W0 / R3

Exactly—19! Nice job. So the rule is “subtract 19,” because 36 − 19 = 17, 35 − 19 = 16, and 34 − 19 = 15. Go ahead and choose D.

Evidence: Go ahead and choose D.

Rationale: The rule and all checks are supplied; selecting the identified option is answer repetition.

Human tutor | E03-B | NA / R3

Yeah. So that's why the rule is to subtract nineteen, 'cause you do it for all of them.

Um, just erasing these so it doesn't get in the way of the next question.

Okay.

Now, what is the answer to this problem? And instead of giving you numbers, they've represented this as blocks. So what is the answer?

Evidence: instead of giving you numbers, they've represented this as blocks

Rationale: A new blocks problem is requested, but neither its operation nor its arrangement is present in preceding text; its demand level cannot be established. The old subtraction rule is correct.

GPT-5.5: explicit scaffolding and rigor instructions | E03-C | W1 / R3

Yes—19. So the rule would be **subtract 19**.

Before we hit check, let’s prove it with another row:

35 − 19 = ?

Evidence: **35 − 19 = ?**

Rationale: After giving the rule, the tutor asks for an unsupplied row calculation in the same table.

E04

Recent conversation

student (T148): One. C.

tutor (T149): Okay, so what is the value of the digit in the tens place?

tutor (T150): [PAUSE: 5 seconds] Student's video feed shows a multiple-choice question on a pink background.

tutor (T150): So what number is in the tens place?

tutor (T151): [PAUSE: 22 seconds]

student (T151): Um.

tutor (T152): So we have ones, tens, hundreds, thousands. So what number is in the tens place?

tutor (T153): [PAUSE: 26 seconds] Student's video feed goes black, then shows a loading icon.

tutor (T153): [SCREEN_UPDATE] Tutor's video feed is turned off.

tutor (T153): [PAUSE: 17 seconds]

student (T153): Four plus one equals five.

tutor (T154): [SCREEN_UPDATE] A notification appears at the top of the screen: "Jordan has left". The student's video feed disappears.

tutor (T154): [PAUSE: 133 seconds] The whiteboard shows the text "thousand hundred eight" and "neigh" in black. At the bottom, "PEMDAS 1" is written in pink and black.

tutor (T154): [SCREEN_UPDATE] A pop-up notification appears: "Jordan has joined".

tutor (T154): Are you able to hear me? It kicked me out.

tutor (T155): [PAUSE: 3 seconds]

student (T155): Oh.

tutor (T156): I'm sorry about that. My whole computer kicked me out of the website.

tutor (T157): [PAUSE: 2 seconds]

tutor (T157): Did you were you able to find the answer for this one?

student (T158): I couldn't see anything. When you go, it goes.

tutor (T159): Okay, I'll give you some time to-

student (T160): S- seven equals twelve.

student (T161): So there's a one that goes there, but there's no four. So is it five?

GPT-5.5: explicit scaffolding and rigor instructions | E04-A | W1 / R1

You’re thinking about carrying — that’s a good start.

Let’s slow down and do just **one column**:

• If a column adds to **12**, you write down the **2**

• Then you carry the **1** to the next column

• If there’s no digit in a spot, that means it’s like adding **0**

Can you read me the two numbers you’re adding, or tell me the digits in the column you’re looking at right now?

Evidence: tell me the digits in the column you’re looking at right now?

Rationale: The learner identifies visible operands or digits; no calculation or explanation is yet required. Carrying rules are valid, though the current displayed problem is unclear even in full context.

Human tutor | E04-B | W1 / R2

So yeah, you said seven plus five is twelve, right? The one carries over right here on top. Two plus one, three. And then what's the bottom number right here?

Evidence: what's the bottom number right here?

Rationale: Identifying a displayed bottom digit is clear work, but the referenced column and two-plus-one step cannot be checked from the incomplete screen description.

GPT-5.5: general tutoring instructions | E04-C | W0 / NA

You're on the right idea with carrying! Let’s slow it down:

For **715 + 555**:

• Ones: **5 + 5 = 10** → write **0**, carry **1**

• Tens: **1 + 5 + 1 = 7** → write **7**

• Hundreds: **7 + 5 = 12** → write **2**, carry **1** to the thousands

So the answer is **1,270**.

And if there’s “no number” in a place, we treat it like **0**, not 4. So it’s not 5 here. The sum is **1,270**.

Evidence: The sum is **1,270**.

Rationale: The complete 715+555 calculation is supplied with nothing left to do. It is numerically correct, but full context does not establish that this earlier sum is the active ambiguous place-value problem at the cut.

E05

Recent conversation

tutor (T212): It is.

tutor (T213): But here we have four four sixes.

student (T214): Oh.

tutor (T215): [PAUSE: 2 seconds] Tutor writes "24" in the first empty box, so the equation reads " = 30 + 24".

Tutor writes "54" in the second empty box, so the equation reads " = 54".

student (T215): Twenty-four.

tutor (T216): Mhm.

student (T217): And when we add that, we get our fifty-four.

tutor (T218): Fifty-four.

student (T219): Yeah.

tutor (T220): So, they just wrote it in a different way, but they did the same thing that we did. They just broke up the numbers.

tutor (T221): [SCREEN_INTERACTION: tutor - editing numbers] Tutor circles "24" and "54" on the screen.

A green checkmark appears on the screen with the text "Great work! You got it. Onward!".

tutor (T221): We're gonna hear the... Okay.

tutor (T222): [PROBLEM_CHANGE: p2 - '9 x 7 = (10 x 7) - (1 x 7)' (start)] Tutor clicks "Next question". The screen changes to a new problem. The problem is "Complete the equations. 9 x 7 = (10 x 7) - (1 x 7)". Below this is " = 70 - [ ]" and " = [ ]".

Tutor clears the previous scratch work from the whiteboard.

tutor (T222): This one's a bit trickier. Okay, let me clear the screen. Here we go.

tutor (T223): So, let's try to understand what an online learning program is doing here, how they set up the problem.

student (T224): Okay.

tutor (T225): [PAUSE: 4 seconds]

tutor (T225): So what do you notice?

student (T226): I notice that the question is harder.

tutor (T227): I noticed that too, but what can you see in the question?

student (T228): Ten times seven minus one times seven.

tutor (T229): [PAUSE: 2 seconds] Tutor writes "7" in the first empty box, so the equation reads " = 70 - 7".

tutor (T229): And why would they do ten times seven minus one times seven?

student (T230): Because ten times seven is seventy and one times seven is seven.

GPT-5.5: explicit scaffolding and rigor instructions | E05-A | W1 / R2

Great — that explains the two pieces. Now connect it back to the original problem: if you have **10 groups of 7** and you take away **1 group of 7**, how many groups of 7 are left?

Evidence: how many groups of 7 are left?

Rationale: The response specifies subtracting one group from ten groups. The learner calculates nine groups; the representation and operation are already supplied, and no explanation is requested.

Human tutor | E05-B | W0 / R0

Okay, so if we did, if we did that like you said, seventy, ten times seven is seventy and one times seven is seven.

Evidence: ten times seven is seventy and one times seven is seven

Rationale: The tutor repeats the learner's two correct products without requesting a further mathematical action.

GPT-5.5: general tutoring instructions | E05-C | W1 / R3

Exactly! Ten sevens is one too many, so they subtract one seven to get back to nine sevens.

So now we have:

70 − 7 = ?

What do you get?

Evidence: 70 − 7 = ?

Rationale: The requested reason for the decomposition is supplied, but subtracting to obtain the final product remains.

E06

Recent conversation

tutor (T12): [SCREEN_INTERACTION: tutor - underlining] Tutor underlines the text "Adding and subtracting fractions".

Tutor types "1. cases: for like fractions" on the whiteboard.

tutor (T12): So first, I'm going to introduce you what's the technique. We're going to watch a video to review the method and then we're going to go over the the practical exercises. Okay, so first let's go over the easiest case that we can see when you want to add or subtract fractions.

tutor (T13): So the first scenario is, let me type for fast. So a first case is for like fractions.

tutor (T14): Like fractions, once again, let me call back the definitions for those. Those are the fractions that have the same denominators. So for example, I could say that we have the fraction three over five. And, uh, hello Nadia.

tutor (T15): [SCREEN_INTERACTION: tutor - writing] Tutor writes "3/5" on the whiteboard.

Tutor writes "4/5" next to "3/5". He then circles the "5" in both fractions.

tutor (T15): Um, another fraction. Okay, hopefully you're doing well. Um, another fraction is four over five. So these two particularly are like fractions because once again, the denominator of these fractions are the same, which is five.

tutor (T16): So let's comes to the operation that we can do for this. So for example, is that these two are like fractions. So for example, I want to add these two fractions, okay? So the thing we need to do is really simple. Hopefully you have you still remember because this is the easiest case that we can learn.

tutor (T17): [SCREEN_INTERACTION: tutor - writing] Tutor writes a plus sign between "3/5" and "4/5".

student (T17): Okay.

tutor (T18): For this case, you just barely need to add the numerator or the above numbers.

tutor (T19): [SCREEN_INTERACTION: student - drawing] Student draws a line above the numerators 3 and 4, connecting them with a plus sign.

tutor (T19): So that would equals to three plus four. Oh, you want to, do you want to do the rest? I can give you some spaces because I believe you already remember what's the technique of doing this.

tutor (T20): [SCREEN_INTERACTION: student - writing] Student writes "= 7/10" as the answer.

tutor (T20): [PAUSE: 3 seconds] Tutor crosses out the "10" in the fraction 7/10 and writes a "5" below it.

Tutor finishes typing "for unlike fractions" under "2. case:".

Tutor writes "5" over a line. He starts to write "3" underneath, then erases it.

Tutor writes "5/3" and "7/4" on the whiteboard.

Tutor draws a double-headed arrow between the denominators 3 and 4. He then draws a "not equal" sign (≠) over the arrow.

Tutor types "These are UNLIKE FRACTIONS" below the fractions.

Tutor erases the "not equal" sign and the arrow. He then writes a plus sign between the two fractions, making the problem "5/3 + 7/4".

Tutor draws a curved line from the numerator 5, crossing the plus sign, down to the denominator 4. He then writes a small "x" for multiplication on the line.

Tutor draws a curved line from the denominator 3, crossing the plus sign, up to the numerator 7. He places dots on the 5, 4, 7, and 3.

Tutor connects the denominator 3 to the numerator 7 with a curved line.

Tutor draws a curved line connecting the two denominators, 3 and 4. He then draws a curved line connecting the numerator 7 to the denominator 3. He adds a multiplication symbol "x" to this line.

Student moves their cursor to the right side of the screen and draws two curved lines resembling parentheses.

Tutor's cursor points to the plus sign inside the butterfly diagram.

Student draws a large, rough, brown U-shape to the right of the problem.

Student continues the U-shape, closing it to form a rough circle or oval. Student then starts writing a brown "M" inside the shape.

Tutor erases the student's brown drawing. Tutor writes "5 x 4" in red to the right of the equals sign.

Tutor writes "3 x 7" in red next to "5 x 4". Tutor then adds a plus sign between "5 x 4" and "3 x 7".

Tutor draws a green fraction bar under "5 x 4 + 3 x 7". Tutor scrolls the whiteboard up to show a previous problem. Tutor then scrolls back down.

Tutor writes "3" and "4" below the green fraction bar, then erases them. Tutor then writes "3 x 4" in red below the green fraction bar.

Tutor writes an equals sign. Tutor then writes "20 + 21".

Tutor draws a red fraction bar under "20 + 21" and writes "12" below it.

tutor (T20): Okay. So looks like you got the right num numerator for the new fraction, but in terms of the denominator, let me tell you that the denominator is kind of special in in such a way, it won't be changed. So basically, when you add two fractions, two like fractions, the the new denominator won't be added, won't be subtracted, won't be anything because these two are the same.

tutor (T21): when you add two fractions, two like fractions, the the the new denominator won't be added, won't be subtracted, won't be anything because these two are five, so we're just gonna copy five and put it exactly right here.

tutor (T22): So, let me tell you that this is actually seven over five, not seven over ten. Um, it's totally fine for you to make mistakes because this is this is uh you just follow a common sense, so it's totally fine for doing so.

tutor (T23): So once again, just remember just remember that whenever you want to add or subtract like fractions, let me go on another example. Numerators are changed. These they are going to be either added or subtracted based on the opera based on the operator, but for the denominator, it won't be changed.

tutor (T24): Okay. So let's go over another example. We can get Okay, so recalling that let me get rid of headlines for this. Give it um a special color. Let's say we have five. We have five. This is also five. So that's this won't be changed.

tutor (T25): So so basically, keep that in mind. Keep that strongly in your mind that these are the rules that you need to follow.

tutor (T26): Okay. So the second scenario, actually the um second operator, excuse me, is that we're going to subtract. And for such I think you just basically do the same as what we we have to do with addition. So for example, I have five over seven and I want to subtract subtract that with by three over seven. Okay, Mateo, uh I'm

going to give you some space for doing this. Do you know how to do this anyway?

tutor (T27): Do you know?

tutor (T28): Or you're not sure for it?

student (T29): Okay.

GPT-5.5: explicit scaffolding and rigor instructions | E06-A | W2 / R2

Okay—go ahead and try it.

Since the denominators are both 7, we **keep the 7**.

So think of it like:

\[

\frac{5}{7} - \frac{3}{7} = \frac{5-3}{7}

\]

What should the numerator be? And why do we keep the denominator as 7?

Evidence: why do we keep the denominator as 7?

Rationale: Beyond 5−3, the learner must explain the unchanged denominator; the rule is stated but its mathematical reason is not.

GPT-5.5: general tutoring instructions | E06-B | W1 / R2

Great. Since both fractions have the same denominator, **7**, we keep the denominator the same.

So:

\[

\frac{5}{7} - \frac{3}{7} = \frac{5-3}{7}

\]

What is **5 - 3**? Can you write the final answer?

Evidence: What is **5 - 3**?

Rationale: The operation and denominator are specified; only the numerator calculation and final fraction remain.

Human tutor | E06-C | W1 / R3

You can try to do it or I can I do it if you certain that you know how to do it, just do it and then I I'm going to correct you whenever you have any if you have any mistakes, otherwise, congrats that you get the right answer.

So just follow the rule we did for um addition. Subtraction happen the same thing. Instead of adding the numerators, you just barely need to subtract those. So these two going to be subtracted.

And the denominator, once again, is the same. So on the right hand side, the denominator is going to be seven. But for the numerator, um maybe let me do it for you as well. So five minus three. It's going to be five minus three over seven. Sorry, the the minus sign is pretty bad. Let me let me let me write it more neatly. Okay, that is good. So this is going to be two over seven.

Hopefully that makes sense. This this is the easiest scenario. I will I believe that and in such case, um it would be even easier than dividing and multiplying fractions.

But barely in the other scenario, you have to do you have to do tons of more work. And that's going to be the second case.

This case is that are these like fraction anyway? Oh yeah, those are like fractions. Um, excuse me, Mateo. I'm I'm not pretty sure if I call this the correct name because I I don't remember like fraction is not

the fractions that have the same denominator. Let me a like denominator. Give me a second. Um what's like fraction? Oh, like fraction is the correct. So, excuse me for that.

I thought like fractions are like follow the same the other definition, not like this. But hope but thankfully it is. So basically, like fraction is a case that denominators between fractions are the same. So two fractions have the same denominators or the bottom numbers are like fractions.

The second case is kind of tricky. You have to add unlike fractions. Unlike fractions, once again, is the trickiest part as I said before.

And it gonna involves you, um, a very really interesting figure, which is a fish as if you still remember.

So, I think I would go over um, exercises rather than just give you the general formula. So, let's get to one exercises.

But unlike fraction. So, let me propose an unlike and two unlike fractions. First, for example, we have, um, five over three. And we're gonna add with, oh shoot, shoot, shoot.

We need to identify first as whether these are unlike fractions or like fractions first. Five over three and seven over four.

Okay, so once again, these two, um, two fractions are either like or unlike. If it is un- if it is unlike, it couldn't be like.

Hopefully that makes sense to you. So, in other words, you have to compare the denominators to see as whether those are like or unlike fractions. And you can see that three is different from four, so those are not equal.

So we could conclude that these are, these are unlike fractions.

Okay, so keep that strongly in your mind. Excuse me for this. Well, the trickiest formula when that you're gonna learn when

you want to add or subtract fractions is that... because we've done the comparison, let me, let me erase this. And then we need to... for example, for example, we want to add these together.

So what's are we going to do is that we just barely follow this figure.

You're gonna start with... Here's the first fraction that we have. So, pay close attention to the above number or the numerator. We're gonna draw the line that gonna cross the addition sign, the plus sign, gonna cross and gonna goes to the denominator of the second fraction. So, these two gonna gonna tell us first we need to do multiplication.

Secondly, we gonna do, move to this. So, we're gonna move at this point. Okay, so let me give you the the dots. Here's the first dot. Here's the first point. Here's the one end point. We have another end point, which is which locates at the the numerator of the second fraction.

So these two, we gonna connect. Um, actually we just draw like this, we won't do anything else. And this the final end point of this is we have... So, basically right now we have three end points. And the the only number which hasn't have any end points is number three over here, which is the denominator. Let me speak more slowly, excuse me. Um, which is the denominator of the first fraction. So give it another end point.

And it's it's going to be the final end point of this figure. Draw like this. And the connection between seven and three is that you're gonna do multiplication.

Pay close attention to this because it's looks kind of intimidating. So as you can see, um, one of the interesting part is that we've done two multiplication. And as you can see this is this is a figure of the fish. If you draw like this, this is the, this is, for example, the eyes of the fish. This is the fin of the fish. We draw like this, but anyways, just trying to figure out that it's going to be a fish.

And this is its head. And inside its head, we see the addition sign.

So what does this means is that once we're done with multiplication, we're gonna add the result together.

So in the loop, we have, or in the head of the fish, we have the addition sign, and that's why when we need to do addition between...

once we multiply the numbers, we're gonna add add those together.

So, practically, this is how it's gonna look like. First, we're gonna start here, five from the endpoint five, we go to endpoint four, connect with multiplication. So, it's going to be five times four.

And then, and then with once, uh, I told you that when we move to four to seven, we don't have to do anything else, so just let it be. But when we, as we go from seven to three, we need to do another another multiplication. So, once again, please close attention that multiplication sign appears at the bottom and then at the top. So, we're gonna have another three mul- uh, multiply seven. Okay.

And these two are separate numbers. So, basically, there's supposed to be something that we need to do with these numbers. We're gonna do either subtracting or any any operations that we need to do. And the signal for this is that inside the loop right here, we have the addition sign, which which tells us that we need to add these numbers.

Hopefully that makes sense. And that's the first step that we need to do with the numerators. For the denominators, it's going to be a little special that these are unlike fractions. So, previously, when you see five, five over here, we're gonna write five immediately on the right hand side. We see seven and seven over here, we're gonna write seven on the right hand side.

But in this case, these three and four are not um, identical. So the, it's going to be a debate that we can't, we can't copy three, we we couldn't put three, we couldn't put four because we have two options. And the method that we need to that you need to remember is that we don't, we won't um, neither write three or four, we're gonna write its product. In other words, we're gonna write three times four as the as the denominator of the new fraction.

So this is the rule that you need to remember, so just keep that in your mind. So that is everything that you need to do to get the new fraction. And to get you master this, I'm gonna give you um, many exercises or we just regularly just do the role play so we can get to everything as we um, supposed to do.

Okay, so I'm gonna do the rest of the calculation. So this is better the methods, the technique. The rest of of those is just really the basic calculation. So, as you remember, five times four is going to be twenty. Recall the multiplication table, which is um, really essential or it's really helpful for you to do these fast. So, five times four is twenty, three times seven is twenty-one. Uh, let me write the the sign right here as how it has been.

So just a little emphasis is that the operator inside the loop is addition and that's why we need to connect this number using addition sign. Three times four, hopefully you still remember, is going to be twelve.

And that's it. It's going to be twenty plus twenty-one. Do you know the the the result of this? Twenty adds twenty-one, twenty adds twenty-one, I mean.

Evidence: Twenty adds twenty-one, twenty adds twenty-one, I mean.

Rationale: The old subtraction is completed, then a different unlike-fractions problem is worked to (20+21)/12. The learner only adds 20+21.

E07

Recent conversation

tutor (T582): [PAUSE: 14 seconds] Tutor writes a 7 to the right of the problem.

student (T582): Twenty-two times.

tutor (T583): Seven can go into twenty-five twenty-two times?

student (T584): Yes, cuz after that number would be twenty-nine.

tutor (T585): Twenty-nine. Hmm.

tutor (T586): [SCREEN_INTERACTION: tutor - writing] Tutor underlines the 25 in the dividend.

Tutor writes "7 R 22" and then crosses it out. He then erases the "R 22".

tutor (T586): Let's see. Just to make sure I understand what you're saying. So you're saying seven can go into twenty-five, twenty-two times.

tutor (T587): [PAUSE: 3 seconds]

student (T587): No. Sorry. Um, I was wrong. I was adding. Sorry.

tutor (T588): You're good.

tutor (T589): [PAUSE: 3 seconds]

tutor (T589): Go through your multiples of seven.

tutor (T590): [SCREEN_INTERACTION: tutor - writing] Tutor writes "21" below the "25".

tutor (T590): [PAUSE: 2 seconds]

student (T590): Okay, twenty-one, sorry.

tutor (T591): Twenty-one, good.

tutor (T592): [PAUSE: 2 seconds]

tutor (T592): And what goes on the top?

tutor (T593): [SCREEN_INTERACTION: tutor - writing] Tutor writes a 3 above the 5 in the dividend, then draws a subtraction line below the 21.

student (T593): Three, sorry.

tutor (T594): Good.

student (T595): Sorry, that is all.

tutor (T596): [PAUSE: 2 seconds]

student (T596): Three.

GPT-5.5: general tutoring instructions | E07-A | W1 / R2

You're good—no need to apologize. Three is correct.

So we do: $3 \times 7 = 21$, then $25 - 21 = 4$.

Now bring down the 6, so we have 46.

How many times does 7 go into 46?

Evidence: How many times does 7 go into 46?

Rationale: The remainder and next partial dividend are supplied; the learner computes the next quotient digit. Full context confirms 2569÷7.

GPT-5.5: explicit scaffolding and rigor instructions | E07-B | W1 / R2

You're good — no apologies needed.

So we have:

• 7 goes into 25 **3** times

• 3 × 7 = **21**

Now subtract: **25 − 21 = ?**

Then we bring down the next digit from 2569, which is the **6**. So after you subtract, tell me what number you get next.

Evidence: Now subtract: **25 − 21 = ?**

Rationale: The learner calculates the remainder and brings down the specified 6, following the supplied division sequence.

Human tutor | E07-C | W1 / R2

Seven into forty-six. So, you're gonna subtract this out, right?

Evidence: you're gonna subtract this out, right?

Rationale: In the full long-division context, 25−21 remains an executable specified subtraction; the next partial dividend 46 is already stated.

E08

Recent conversation

student (T68): I got twelve percent.

tutor (T69): Not okay, so how did you get that? What did you do first?

student (T70): Oh, we're not supposed to divide or?

tutor (T71): No.

tutor (T72): Well, what were you dividing?

tutor (T73): Which two numbers?

student (T74): Oh wait.

student (T75): So I divided seventy-two divided by six.

tutor (T76): Perfect.

student (T77): And I did twelve times six, which gave me seventy-two.

tutor (T78): Twelve times. Okay, so you're on the right track.

tutor (T79): That is perfect. So what you were doing is totally right. So you do 72 divided by six, and what is that telling you? That's saying in how many days he cleans 12 rooms. If you do 72 divided by six.

student (T80): Oh, I did not see that. How many rooms can ribbon- oh no, Robin clean in nine days?

tutor (T81): So now what do you have to do?

student (T82): After I do um, 12 divided by nine?

tutor (T83): Not quite. Okay, you're really close, but before you were finding the rate because you wanted to go from six days to one day, so you divided. But now you want to go from your one day, which is 12 rooms, to nine days. So if I'm trying to get to a bigger number, should I divide or multiply?

student (T84): Multiply. Yeah, for a bigger number.

tutor (T85): Mhm. Perfect.

tutor (T86): [PAUSE: 11 seconds]

student (T86): I got my answer.

tutor (T87): All right, what do you got?

student (T88): One hundred and eight percent. Unless that's not how you do- yeah.

tutor (T89): Percent?

student (T90): One hundred and eight percent.

GPT-5.5: explicit scaffolding and rigor instructions | E08-A | W1 / R3

The number is right — 108 — but not percent this time.

Look at what the question is asking: “How many **rooms** can Robin clean in 9 days?” So the unit is **rooms**, not percent.

So we'd say: **108 rooms**.

Can you tell me the rate sentence out loud? Like: "Robin cleans ___ rooms per day."

Evidence: Robin cleans ___ rooms per day.

Rationale: The final unit and answer are supplied. The learner retrieves the previously found 12 as the rate in a strongly cued sentence.

GPT-5.5: general tutoring instructions | E08-B | W0 / R3

Yes—108 is right! Just not percent this time. It would be **108 rooms**.

Because:

• 72 ÷ 6 = 12 rooms per day

• 12 × 9 = 108 rooms in 9 days

So type **108**.

Evidence: So type **108**.

Rationale: Both calculations and the rooms unit are supplied; only entering the answer remains.

Human tutor | E08-C | W0 / R0

One hundred and eight? Or

Evidence: One hundred and eight? Or

Rationale: The fragment requests clarification or repetition of the learner's number, with no new mathematical operation or definite correction.

E09

Recent conversation

tutor (T326): Mhm.

student (T327): Because it only has one way to make five.

tutor (T328): And it has five. Uh-huh.

student (T329): And it has two um, what's it called? It has two numbers, so that means it's a um, a prime number.

tutor (T330): [SCREEN_UPDATE] The YouTube video narrator says, "is a positive integer."

tutor (T330): Oh, okay. Okay.

tutor (T331): [SCREEN_UPDATE] The YouTube video narrator says, "Three is a positive integer, and when you do five multiplied by three, you get fifteen. So both five and three are factors of fifteen."

tutor (T331): [PAUSE: 6 seconds]

tutor (T331): [SCREEN_UPDATE] In the "Thinking" column on the whiteboard, the text "Why is 15 not prime?" appears.

tutor (T331): [SCREEN_UPDATE] The YouTube video narrator says, "Why is fifteen not a prime number?"

tutor (T331): So we just talked about that one already. So let's see what it has to say.

tutor (T332): [SCREEN_UPDATE] The YouTube video narrator says, "Fifteen has four factors. Prime numbers have exactly two factors, so fifteen is not prime."

tutor (T332): [PAUSE: 5 seconds]

tutor (T332): [SCREEN_UPDATE] In the "Thinking" column on the whiteboard, the text "Why is 1 a factor?" appears.

tutor (T332): [SCREEN_UPDATE] The YouTube video narrator says, "a factor of seventeen... integer. Seventeen is a positive integer... teen, we get an of seventeen."

tutor (T332): [SCREEN_UPDATE] In the "Thinking" column on the whiteboard, the text "Why is 7 not a factor?" appears.

tutor (T332): [SCREEN_UPDATE] The YouTube video narrator says, "Why is... of seventeen... a positive integer, but there's no to get an not a factor of seventeen."

tutor (T332): [SCREEN_UPDATE] The YouTube video narrator says, "Why is seventeen a prime number?"

tutor (T332): [PAUSE: 5 seconds]

tutor (T332): [SCREEN_UPDATE] The YouTube video narrator says, "Seventeen has exactly two factors. And prime numbers have exactly two factors."

tutor (T332): Okay. All right, Lily, where did you go? I think I have lost you in math. Okay.

tutor (T333): Hopefully this is not really boring, but at the end of now we're at the end of the video. Okay.

tutor (T334): So, what's the different or do you understand what factor and prime kind of mean or is it kind of still a little confusing?

student (T335): Um, the factor is confusing.

Human tutor | E09-A | W0 / R3

The factor? That's okay. So, basically, I'll kind of explain it in a more simple way too, or another way.

Factor is kind of like, you see how, let's put seventeen as an example, right? So you see how seventeen, it says factors of seventeen is one times seventeen. Factors basically means if you got to think of like a, like kind of like a family tree, right? Where you have two, you have like, you know, two, two numbers and it makes up an answer, right?

So factors basically means what are the um, what are the, fact, kind of like what are the numbers that makes up an answer?

Evidence: what are the numbers that makes up an answer?

Rationale: The rhetorical definition question is answered by the explanation itself; no specific unsupplied factor judgment is requested. The multiplication framing is valid though imprecise.

GPT-5.5: general tutoring instructions | E09-B | W0 / R3

No worries — factors can be confusing at first. Let’s slow it down.

A **factor** is a number that can multiply with another whole number to make the number.

Example: for **12**:

1 × 12 = 12

2 × 6 = 12

3 × 4 = 12

So the factors of 12 are: **1, 2, 3, 4, 6, 12**.

Another way to think about it: a factor divides evenly with **no leftovers**.

So **12 ÷ 3 = 4**, no leftovers, so **3 is a factor of 12**.

But **12 ÷ 5** has leftovers, so **5 is not a factor of 12**.

Let’s try one together: Is **2** a factor of **12**? Why or why not?

Evidence: **2 × 6 = 12**

Rationale: The new example's factor 2 and its justification have both already been supplied before asking whether and why it is a factor. Repeating that example remains.

GPT-5.5: explicit scaffolding and rigor instructions | E09-C | W0 / R3

No worries — thank you for telling me. Let’s pause and make “factor” less confusing.

Think of a **factor** as a number that can fit evenly into another number — no leftovers.

For example, with **12**:

• **3 is a factor of 12** because **3 × 4 = 12**

• **5 is not a factor of 12** because there’s no whole number times 5 that equals 12

So the question is always:

Can I multiply this number by another whole number to get the big number?

Let’s try one together:

Is **4** a factor of **12**?

What number could go with 4 to make 12?

Evidence: **3 is a factor of 12** because **3 × 4 = 12**

Rationale: The new example's 3×4=12 already supplies both 4's factor status and its partner 3. The final two questions only ask the learner to identify information explicitly given in that equality.

E10

Recent conversation

tutor (T254): Yeah. So minus fifty-three. So, what do you think eighty-two minus fifty-three is?

tutor (T255): [SCREEN_INTERACTION: tutor - scribbling] Tutor scribbles over the answer "29" on the right side of the whiteboard to hide it.

tutor (T255): [PAUSE: 16 seconds] Tutor writes "29" under the subtraction problem on the left.

Tutor writes "82 - 53" and "39 almost" on the whiteboard.

Tutor writes out the subtraction problem "82 - 53" with borrowing, showing the 8 becoming a 7 and the 2 becoming a 12 (represented as a 10). She makes a few calculation errors while talking through it but ends with the correct answer of 29.

tutor (T255): And you can say whatever you're thinking. Don't worry.

student (T256): Twenty-nine.

tutor (T257): Twenty-nine. Yes.

tutor (T258): Did you count with your fingers to get twenty-nine? Mhm. Yeah, so that's good. Yeah, cuz I

tutor (T259): like how you got the correct digit, right? The nine. But then you also did the um carryover. Right?

tutor (T260): So what I mean by that is like eighty-two minus fifty-three, if we just did it like this, if you got this, which this would be almost correct, right? But then you kind of recognized and then you did ten minus three, which is seven. Seven plus two, which is nine. Turn this eight to a seven. Seven minus three, which is four. Right? Oh sorry, my bad. Which is a two. Sorry, I don't know why I was reading three, five. Which is two. Yes, so twenty-nine.

tutor (T261): [SCREEN_INTERACTION: tutor - typing] Tutor types "29" into the answer box for the practice question.

tutor (T261): [SCREEN_UPDATE] The answer is submitted and a feedback screen appears showing "Your answer: 29".

Tutor clicks the "Yes, watch video anyway" button.

tutor (T261): Yay. Or my bad. I forgot all the units. Twenty-nine dollars. Sorry, I'm so used to not doing it. So, we'll watch it real quick.

tutor (T262): [SCREEN_UPDATE] A video player appears and starts playing. The video shows the problem text.

tutor (T262): [SCREEN_UPDATE] The video narrator explains how to solve the problem by first subtracting 45 from 82, and then subtracting 8 from the result (37), to get the final answer of 29. The calculations are shown on the screen.

Tutor draws an arrow from the 37 to the 29 and writes "82 - (45+8)" on the screen over the video.

tutor (T262): And what you did the before, the eighty-two minus forty-five plus eight is correct too. They just didn't show it here.

tutor (T263): Okay.

tutor (T264): [SCREEN_UPDATE] The video ends and the tutor clicks the "Let's do it!" button.

tutor (T264): Okay. So let's do this one. We'll do maybe two, three more if that's good.

tutor (T265): [PAUSE: 1 second]

tutor (T265): So, Frank has six, sorry, 8.6 centimeters of glue stick. He uses 17 millimeters to glue a model. How much does he have left in millimeters?

tutor (T266): Hm, interesting.

tutor (T267): So, do we remember what we said about the units?

tutor (T268): [PAUSE: 4 seconds] Tutor writes "1 cm = 10 mm" on the whiteboard.

Tutor writes "1.0 cm -> 10 mm" and "8.6 cm = 86 mm" on the whiteboard.

student (T268): No.

GPT-5.5: explicit scaffolding and rigor instructions | E10-A | W1 / R2

That's okay — this one has a new unit, so let's slow it down.

A **centimeter** is the bigger unit, and a **millimeter** is smaller.

So:

1 centimeter = 10 millimeters

So if Frank has **8.6 centimeters**, we can think of that as:

• **8 centimeters = 80 millimeters**

• **0.6 centimeters = 6 millimeters**

So altogether, **8.6 cm = 86 mm**.

Now the problem is asking: he starts with **86 millimeters** and uses **17 millimeters**.

So what subtraction problem should we do now?

That's okay — this one has a new unit, so let's slow it down.

A **centimeter** is the bigger unit, and a **millimeter** is smaller.

1 centimeter = 10 millimeters

So if Frank has **8.6 centimeters**, we can think of that as:

• **8 centimeters = 80 millimeters**

• **0.6 centimeters = 6 millimeters**

So altogether, **8.6 cm = 86 mm**.

Now he starts with **86 millimeters** and uses **17 millimeters**.

What subtraction problem should we do now?

Evidence: What subtraction problem should we do now?

Rationale: The learner identifies 86−17 after being given both converted quantities and the subtraction cue; no open strategy choice remains.

GPT-5.5: general tutoring instructions | E10-B | W1 / R2

No worries—that’s okay. This one is about changing units first.

Centimeters are bigger than millimeters:

1 centimeter = 10 millimeters

So **8.6 cm** means **8 centimeters and 6 millimeters**.

8 centimeters = **80 mm**

plus 6 mm = **86 mm**

So Frank starts with **86 millimeters**.

Now he uses **17 millimeters**, so we need to do:

86 − 17

Can you try that subtraction for me?

Evidence: Can you try that subtraction for me?

Rationale: The conversion and 86−17 expression are supplied; computing 69 millimeters remains.

Human tutor | E10-C | W0 / R2

I'll it's okay. I'll I'll show you. So basically,

one centimeter equals ten millimeters. And what does this mean? Basically, if we have 1.0 centimeters, if we want to turn it to millimeters, we're going to just move this one over here. So this will become ten. And I'll show you what this means again with this 8.6 centimeters. We're going to move this one over, and then this will turn into eighty-six millimeters.

Does that make sense?

Evidence: this will turn into eighty-six millimeters

Rationale: The unit question is explained and answered, ending only in an understanding check; the final subtraction is not elicited. The earlier 'what does this mean' is self-answered.

E11

Recent conversation

tutor (T181): Yeah, good job.

tutor (T182): Okay. So it looks like you got the hang of it.

tutor (T183): So let's do this on an online learning platform now.

tutor (T184): [SCREEN_UPDATE] The whiteboard is replaced by a black screen with the text "dq4bJ9OCXnLS5luyacUe".

tutor (T184): [PAUSE: 1 second]

tutor (T184): Okay.

tutor (T185): So the first question's asking you to convert 0.17 to a percentage. So how would you do that?

student (T186): Um, do 0.17 times 100.

tutor (T187): Mhm. And what would that be?

tutor (T188): [PAUSE: 2 seconds]

tutor (T188): [PROBLEM_CHANGE: p1 - 'Convert this decimal to a percentage' (start)] A problem appears on screen. "Convert this decimal to a percentage" with "0.17" below it. Four multiple choice options are shown: A) 0.17%, B) 1.7%, C) 17%, D) 170%.

tutor (T188): [PAUSE: 10 seconds]

student (T188): Um, D.

tutor (T189): Not D.

tutor (T190): So try to move the decimal places two two places to the right.

tutor (T191): So move the decimal point two to the right. What would you get?

tutor (T192): [PAUSE: 28 seconds] Tutor switches back to the whiteboard.

Tutor writes "0.17" and draws two arcs from the decimal point to the right.

tutor (T192): Or let's try doing it on the board together.

tutor (T193): So we have zero... oh. We have 0.17 and we want to convert to a percent. So yeah, you were right when you said we have to multiply by 100. So when you multiply by 100, remember you're just moving two decimal places to the right.

tutor (T194): So what do you get when you move it over here?

tutor (T195): [PAUSE: 34 seconds] Tutor writes "= 17."

Tutor switches back to the an online learning platform problem.

tutor (T195): So when you multiply by a hundred, all you get is 17. 'Cause now the decimal place is right there.

tutor (T196): Does that make sense?

student (T197): Yeah.

GPT-5.5: general tutoring instructions | E11-A | W0 / R3

Great. So $0.17 \times 100 = 17$, and since it's a percent, we write 17%.

Which answer choice is 17%?

Evidence: Which answer choice is 17%?

Rationale: The percentage is supplied; the learner only selects its matching displayed option.

GPT-5.5: explicit scaffolding and rigor instructions | E11-B | W2 / R3

Good. So $0.17 \times 100 = 17$, and then we add the percent sign: 17%.

Which answer choice matches 17%? And why do you think it's not 170%?

Evidence: why do you think it's not 170%?

Rationale: Although selecting 17% repeats the answer, explaining why 170% is wrong requires an additional place-value justification.

Human tutor | E11-C | W1 / R3

Okay.

So if we go back to the an online learning platform, our answer would be C.

So yeah, it gives a little explanation.

Now, how about this? Point zero seven to a percentage. Do you know how you would do that?

Okay. Do you know what zero point zero seven...

times one hundred?

Evidence: zero point zero seven...

times one hundred?

Rationale: The old choice is given, and the learner must calculate 0.07×100 for a different decimal.

E12

Recent conversation

tutor (T79): [SCREEN_INTERACTION: tutor - drawing] Circles the topics "Angles at a point" and "Angles on a straight line".

student (T79): This one.

tutor (T80): [PAUSE: 7 seconds]

student (T80): The one above it. Angles on a straight line.

tutor (T81): [SCREEN_INTERACTION: tutor - clicking] Clicks the "Start topic" button next to "Angles on a straight line".

tutor (T81): [PROBLEM_CHANGE: p1 - 'Adjacent angles on a straight line sum to...' (start)] A new problem loads. The question is "Adjacent angles on a straight line sum to...". The options are A: 90°, B: 180°, C: 360°, D: 100°.

tutor (T81): [PAUSE: 6 seconds]

student (T81): Um.

tutor (T82): [PAUSE: 5 seconds] Clicks on option B, then clicks on option C.

student (T82): B.

student (T83): No, actually C.

tutor (T84): [SCREEN_INTERACTION: student - clicking] Clicks the "Check" button.

tutor (T84): [SCREEN_UPDATE] A green banner appears with "C is correct!". The student clicks the "Next" button.

tutor (T84): [PROBLEM_CHANGE: p2 - 'Tom and Katie are discussing adjacent angles on a straight line.' (start)] A new problem loads. It shows two diagrams. Tom's diagram shows two angles next to each other on a straight line. Katie's diagram shows two angles on a straight line but not touching. The question is "Who do you agree with?". The options are A: Only Tom, B: Only Katie, C: Both Tom and Katie, D: Neither is correct.

tutor (T84): Tom and Katie are discussing adjacent angles on a straight line. Tom says these are adjacent angles on a straight line. Katie says these are adjacent angles on a straight line. Who do you agree with?

student (T85): B.

tutor (T86): [SCREEN_INTERACTION: student - clicking] Clicks on option B, then clicks the "Check" button.

tutor (T86): [SCREEN_UPDATE] A red banner appears with "That's incorrect". The explanation reads: "Katie's diagram shows 2 angles that are not adjacent on the straight line, which means they are not touching."

tutor (T86): So what does adjacent mean?

tutor (T87): [SCREEN_INTERACTION: student - clicking] Clicks the "Close" button on the feedback banner.

Draws a straight line, then draws a ray from a point on the line, creating two adjacent angles.

tutor (T87): So adjacent means that they're touching. So see how in Tom's, they're touching?

student (T88): Between?

tutor (T89): So adjacent means that they're touching. What does it mean?

student (T90): What does it mean?

Human tutor | E12-A | W0 / R3

So adjacent means that they're touching. So see how in Tom's, they're touching? So if I have a straight line, and I have an angle here, and an angle here, these two angles are adjacent.

Evidence: So adjacent means that they're touching.

Rationale: The meaning and Tom's example are explicitly supplied; the question only directs agreement with that supplied relationship.

GPT-5.5: explicit scaffolding and rigor instructions | E12-B | W1 / R1

Adjacent means **next to each other and touching**.

For angles, adjacent angles must:

1. share the same corner point, and

2. share one side/ray, so they are right beside each other.

Not just “between” — more like **neighbors**.

Look at Tom’s and Katie’s diagrams: which one has the two angles touching like neighbors?

Evidence: which one has the two angles touching like neighbors?

Rationale: The learner recognizes touching angles under the stated criteria; preceding context already explicitly identifies Tom's touching diagram, making this heavily cued.

GPT-5.5: general tutoring instructions | E12-C | W0 / R3

“Adjacent” means **right next to each other and touching**.

For angles, adjacent angles:

• share the same corner point (vertex)

• share one side/ray

• have no gap between them

So in the picture, Tom’s angles are adjacent because they are touching. Katie’s are not adjacent because they are separated.

Which choice matches that?

Evidence: Tom’s angles are adjacent because they are touching. Katie’s are not adjacent

Rationale: Both diagram judgments are given before asking only for the matching option.

E13

Recent conversation

student (T72): Um, no.

tutor (T73): [PROBLEM_CHANGE: p1 - 'Penny is counting in 10s. What number would she say after 590?' (start)] The problem appears on the screen. The question is "Penny is counting in 10s. What number would she say after 590?". The multiple-choice options are A: 580, B: 591, C: 5100, D: 600.

tutor (T73): Okay. Can you go ahead and read that for me?

student (T74): Penny is counting in tens. What number would she say after five hundred ninety?

tutor (T75): [PAUSE: 2 seconds] Tutor writes "590" on the whiteboard, then "+ 10" below it. He explains that the 0 in the ones place stays, the 9 in the tens place becomes 10, so a 0 goes in the tens place and the 1 is carried over to the hundreds place, making the 5 a 6. The final answer is "600".

tutor (T75): Right. So there's, there's a clear option that's wrong because if you add ten here, there should not be a one in one's place, right?

student (T76): Yeah.

tutor (T77): Okay, so B's wrong. What do you think the right one is?

student (T78): So it's pretty much asking five hundred and nine hundred, five hundred, nine hundred times, times ten?

tutor (T79): No, it's asking five hundred and ninety plus ten.

student (T80): Okay, so plus ten, um, six hundred.

tutor (T81): Yeah, six hundred, right.

tutor (T82): Okay. So the strategy to do this faster is like every time you're given a number and you're adding ten, fastest way to do it is this stays the same. This one increases by one, so it'll be ten, but one goes here and it's six hundred. And then the rest of it should just fall into place. Six hundred's right.

tutor (T83): [PAUSE: 1 second] Student clicks on option D. Tutor clicks "Check". The answer is confirmed as correct.

tutor (T83): Um, but try this, try this approach for the next few questions if you need to.

tutor (T84): [PROBLEM_CHANGE: p2 - 'When skip counting in 10s from 780, which of these numbers would you not say?' (start)] The next problem appears. The question is "When skip counting in 10s from 780, which of these numbers would you not say?". The options are A: 800, B: 790, C: 990, D: 789.

tutor (T84): [PAUSE: 3 seconds] Student clicks on option C.

student (T84): Okay.

tutor (T85): So,

student (T86): then, seven hundred ninety.

tutor (T87): No, no, read the, read the question again.

student (T88): Uh, when skip counting in tens from seven hundred eighty, which of these numbers would you not say?

tutor (T89): Mhm.

student (T90): Oh. Uh, I wouldn't say C.

Human tutor | E13-A | W0 / R2

Let's check it.

Oh, that's wrong.

Evidence: Oh, that's wrong.

Rationale: The response rejects C correctly without eliciting another calculation or explanation; the final answer remains unsupplied.

GPT-5.5: general tutoring instructions | E13-B | W1 / R2

Good catch rereading it. Let’s check C together: if we start at 780 and count by 10s, we get 780, 790, 800, 810… eventually we would get to 990.

So C actually is a number we would say.

When counting by 10s, the ones digit stays the same. Starting at 780, the ones digit is 0, so every number should end in 0. Which option does not end in 0?

Evidence: Which option does not end in 0?

Rationale: The decisive invariant and rejection of C are supplied; the learner recognizes the only displayed option lacking a final zero.

GPT-5.5: explicit scaffolding and rigor instructions | E13-C | W2 / R2

Let’s pause—this is a **“would you NOT say”** question, so we have to be extra careful.

Why do you think **C: 990**? Is 990 impossible to reach by adding 10 over and over from 780, or is it just farther away?

Try counting a few:

780, 790, 800, 810...

What do you notice about the **ones digit** every time? Then compare that to the answer choices.

Evidence: What do you notice about the **ones digit** every time?

Rationale: The learner must infer the invariant from the given sequence and connect it to the choices, also reconsidering the reason for C.

E14

Recent conversation

tutor (T43): [PAUSE: 3 seconds]

student (T43): Because

tutor (T44): [PAUSE: 2 seconds]

student (T44): like

tutor (T45): [PAUSE: 12 seconds]

student (T45): Because I count one, two, three, four.

tutor (T46): [PAUSE: 2 seconds]

tutor (T46): Okay. So you think it's um letter A?

student (T47): Yes.

tutor (T48): Okay.

tutor (T49): [SCREEN_UPDATE] Tutor clicks the "Check" button. The screen shows "Try again". Tutor clicks "Check" again. The screen shows "A is correct! The grasshopper jumps from 0cm to 8cm. 8 + 6 = 14, so it lands 14cm altogether."

tutor (T49): Is this how you did it? We look at the explanation.

tutor (T50): [SCREEN_INTERACTION: student - drawing] Student draws a series of dots on the whiteboard.

student (T50): Because yeah, I'm just gonna want to push next.

tutor (T51): Okay.

tutor (T52): [PROBLEM_CHANGE: p2 - 'A grasshopper starts at 5cm, jumps forward 7cm, and then jumps back 3cm. Where does the grasshopper land?' (start)] Student clicks the "Next" button. A new problem appears. It reads: "A grasshopper starts at 5cm, jumps forward 7cm, and then jumps back 3cm. Where does the grasshopper land?" Below the problem is a ruler from 0 to 20. There are four multiple-choice options: A) 15cm, B) 9cm, C) 4cm, D) 10cm.

tutor (T52): What about this one?

student (T53): Uh, ten. Fifteen.

tutor (T54): Why would you say that? How did you get there?

tutor (T55): [PAUSE: 7 seconds]

tutor (T55): Why don't we do it this way? Okay. So where does it say it starts?

tutor (T56): [PAUSE: 7 seconds] Student writes "5" on the whiteboard.

Student writes "+" next to the "5".

tutor (T56): Where does the grasshopper start?

student (T57): Because I know it if you like if you like this, put it like this. Like... like... five.

Human tutor | E14-A | W0 / R0

So it starts on five, right?

Evidence: So it starts on five, right?

Rationale: This asks only assent to the already stated starting value.

GPT-5.5: explicit scaffolding and rigor instructions | E14-B | W1 / R2

Yes, exactly — the grasshopper **starts at 5**.

Put your pencil or finger on **5** on the ruler. Now let's do only the first jump:

It jumps **forward 7 cm**. Forward means toward the **bigger numbers**.

Can you count 7 spaces from 5 and tell me what number it lands on after that first jump?

Evidence: count 7 spaces from 5

Rationale: The starting point, direction, and number of steps are specified; counting to the first landing remains.

GPT-5.5: general tutoring instructions | E14-C | W0 / R3

Yes! Start at **5**.

So we can write:

5 + 7 - 3

"Forward 7" means we **add 7**.

Let's count from 5: **6, 7, 8, 9, 10, 11, 12**.

Now it jumps **back 3**, so count back from 12: **11, 10, 9**.

So where does the grasshopper land? Which answer choice is **9 cm**?

Evidence: Which answer choice is **9 cm**?

Rationale: Both jumps are fully counted and the landing supplied before the learner selects the matching option.

E15

Recent conversation

tutor (T134): Oh, that's cool. Do you like farming? Are you interested in it? Or do you kind of just have to go?

student (T135): I uh, I like it. And my my cousin goes there too.

tutor (T136): Oh, your cousin goes to it. That's nice.

tutor (T137): [PAUSE: 4 seconds]

tutor (T137): Okay. Um, well your tutor, your actual tutor said that you usually start with a question of the day. Um, I don't know what that is. Do you know what that is?

student (T138): Yeah.

tutor (T139): What's that?

student (T140): It's like somewhere like on a different uh app.

tutor (T141): It's like somewhere on app? Somewhere on an app?

student (T142): Yeah.

tutor (T143): Uh, do you know what app that is?

student (T144): It's a learning platform.

tutor (T145): Okay, on a learning platform. Hold on, let me let me share my screen. Then maybe we could start with that.

tutor (T146): [PAUSE: 30 seconds]

tutor (T146): [SCREEN_INTERACTION: tutor - sharing screen] Tutor shares a screen showing a website.

student (T146): At the very bottom.

tutor (T147): [SCREEN_INTERACTION: tutor - scrolling] Tutor scrolls down the page. A problem is visible. It says "76 = ? - 24".

tutor (T147): Oh, there it is.

tutor (T148): [PROBLEM_CHANGE: p1 - '76 = ? - 24' (start)] The problem on the screen is "76 = ? - 24". Below it are four answer choices: 52, 90, 100, 110.

tutor (T148): So 76 is what minus 24?

tutor (T149): Do you know how to do this one?

tutor (T150): [PAUSE: 1 second]

tutor (T150): [PAUSE: 19 seconds]

student (T150): No.

Human tutor | E15-A | W1 / R2

No?

Okay. So, in order to do this, if we want to subtract 24 from a number, then to do that, we would just take 76 and add 24 to go the opposite direction.

So what happens if we do 76 plus 24? Do you know what that is?

Evidence: what happens if we do 76 plus 24?

Rationale: The inverse operation is supplied; the learner computes the missing initial number.

GPT-5.5: general tutoring instructions | E15-B | W0 / R3

That's okay — let's do it together.

76 = ? − 24

So we're asking: **What number minus 24 gives us 76?**

To undo "minus 24," we do the opposite: **add 24**.

76 + 24 = ?

Let's break it up:

76 + 20 = 96

96 + 4 = 100

So the missing number is **100**.

Let's check: **100 − 24 = 76** ✓

So the answer is **100**.

Evidence: So the missing number is **100**.

Rationale: The response answers its own missing-number questions and completes the check without a remaining demand.

GPT-5.5: explicit scaffolding and rigor instructions | E15-C | W2 / R1

That's okay. Let's think of it like a story:

"I had some number of things. I took away 24. Then I had 76 left."

So the mystery number has to be **bigger than 76** or **smaller than 76**?

Evidence: **bigger than 76** or **smaller than 76**?

Rationale: The learner infers the ordering relationship between the unknown starting quantity and the remainder from the story.

E16

Recent conversation

tutor (T238): Which one should we convert?

student (T239): Oh, that means I have to convert the grams.

tutor (T240): Yep.

student (T241): So then, one kilogram is one thousand grams. So then that means we have to do four hundred seventy-three, but that isn't a thousand.

tutor (T242): So how do you convert grams to kilograms?

student (T243): Decimal. I think it'd be like the number before the decimal.

tutor (T244): What do you mean?

student (T245): So is it like four point seventy-three?

tutor (T246): So if you have one thousand grams, and that equals one kilogram, right? If you have four hundred and seventy-three grams, how many kilograms is that? Is that more or less than one kilogram?

student (T247): Kind of.

tutor (T248): Less.

tutor (T249): So if you have four hundred and seventy-three grams, right? And you wanna convert it to kilograms, you're gonna divide by a thousand. So where does the decimal go?

student (T250): Oh.

tutor (T251): So if you have four hundred and seventy-three, the decimal's right here, right? And you're dividing by a thousand, which has three zeros. So you're gonna move the decimal one, two, three times. So it's gonna be point four seven three.

student (T252): Would it be four like a full circle and that would be the four and then like a seventy-three?

tutor (T253): What do you mean?

student (T254): My whiteboard marker is out. I have to get a piece of paper.

tutor (T255): Okay, get a piece of paper.

student (T256): Okay, so...

tutor (T257): Oh yeah.

student (T258): So, would it be like a, a full circle like this? Right? And then that would be four?

tutor (T259): No.

tutor (T260): So you see how four hundred and seventy-three is less than a thousand? So it's not gonna be a full kilogram. It's gonna be a part of a kilogram.

student (T261): So then it's not a full circle. So then, would we, it'd be like a half, like a, less than half.

Human tutor | E16-A | W0 / R0

Yeah.

Evidence: Yeah.

Rationale: This confirms the learner's correct less-than-half interpretation and requests no additional action.

GPT-5.5: general tutoring instructions | E16-B | W1 / R2

Yes! It’s a little less than half, because half of 1,000 g is 500 g, and 473 g is just under 500 g.

So:

473 g = 0.473 kg

Now add it to 1.8 kg. Write 1.8 as 1.800 so the decimals line up:

1.800 + 0.473 = ?

What do you get?

Evidence: 1.800 + 0.473 = ?

Rationale: Full context confirms the same task is adding 1.8kg and 473g. Conversion and alignment are supplied, leaving decimal addition.

GPT-5.5: explicit scaffolding and rigor instructions | E16-C | W1 / R2

Yes — it’s a little less than half of a kilogram, because half of 1000 grams is 500 grams, and 473 is just under 500.

For the exact decimal: 473 grams is **473 out of 1000** of a kilogram.

So how would you write **473 thousandths** as a decimal?

Evidence: write **473 thousandths** as a decimal?

Rationale: The learner translates a specified place-value quantity into decimal notation; the exact decimal was also given earlier, making the request strongly cued.

E17

Recent conversation

student (T63): I don't know.

tutor (T64): You sure?

tutor (T65): [PROBLEM_CHANGE: p1 - 'What is the result of multiplying 3/4 by 3 ?' (start)] The problem appears on the screen with four multiple-choice options. A: 6/4, B: 9/4, C: 1, D: 9/12.

student (T65): Maybe A.

tutor (T66): [PAUSE: 6 seconds]

student (T66): I said A.

tutor (T67): Not A.

tutor (T68): Eight? There is no eight there, buddy.

tutor (T69): Let's go to the whiteboard.

student (T70): I said A, not...

tutor (T71): Let's go to the whiteboard for this.

tutor (T72): [SCREEN_UPDATE] The screen switches back to the blank whiteboard.

Tutor writes "3/4 x 3/1" on the whiteboard.

tutor (T72): Three over four times three is the same thing as doing three over four times three over one.

tutor (T73): So what's the answer?

tutor (T74): [PAUSE: 6 seconds] Tutor writes an equals sign after the problem.

tutor (T74): You would do three times three over four times one. So what does it equal?

tutor (T75): [PAUSE: 13 seconds]

student (T75): So, four times one equals four and then three times three equals nine.

tutor (T76): [PAUSE: 11 seconds]

student (T76): Four nine.

student (T77): Three plus four.

tutor (T78): [PAUSE: 3 seconds]

tutor (T78): Nine, yes.

student (T79): Four.

GPT-5.5: explicit scaffolding and rigor instructions | E17-A | W2 / R2

Good — you found the two parts: 9 and 4.

Now, which one goes on top in the fraction: the top-times-top answer, or the bottom-times-bottom answer?

So would it be **9/4** or **4/9**?

Evidence: which one goes on top in the fraction

Rationale: The products are known, but the learner must map numerator and denominator products to their positions; the correct arrangement is not stated.

GPT-5.5: general tutoring instructions | E17-B | W0 / R3

Exactly — 3 × 3 = 9 on top, and 4 × 1 = 4 on bottom.

So the answer is **9/4**.

Which letter choice is **9/4**?

Evidence: So the answer is **9/4**.

Rationale: The complete fraction and its positions are supplied; choosing its letter repeats that answer.

Human tutor | E17-C | W1 / R2

Nine over what?

Evidence: Nine over what?

Rationale: The numerator is placed for the learner, who must identify the denominator from the already computed 4×1 product.

E18

Recent conversation

tutor (T185): Oh, I kinda just bought it cuz I wanted one, but I didn't I didn't not know how to use those.

student (T186): Yeah.

tutor (T187): Yeah.

tutor (T188): All right. So...

student (T189): You have a Furby?

tutor (T190): Yes, I did have a Furby.

tutor (T191): Oh my goodness. That just like took me back.

student (T192): I don't know. I never had... Wait, I did have a Furby.

student (T193): Cuz it was like some old...

tutor (T194): [PAUSE: 18 seconds]

tutor (T194): Oh.

tutor (T195): [PAUSE: 5 seconds]

tutor (T195): [PROBLEM_CHANGE: p1 - 'Find the value of the star' (start)] A new problem appears on the screen. It shows a picture of a sheep next to a math problem. The problem is "4 x ★ = 2,284". Below the problem are four multiple-choice options: A) 571, B) 521, C) 2,221, D) 520.

student (T195): It's kind of scary though.

tutor (T196): Yeah, they they kind of are.

student (T197): Yeah.

tutor (T198): Okay. So let's go ahead and get started since it's six thirty.

tutor (T199): Um, so we're gonna do some division today.

student (T200): Okay.

tutor (T201): All right. So, okay.

tutor (T202): So we have two thousand... So the problem says we have four times star equals two thousand two hundred eighty-four.

tutor (T203): So if we want to find the star, what do we have to do?

tutor (T204): [SCREEN_INTERACTION: tutor - writing] Tutor writes "2284" on the board.

Tutor writes the long division symbol around "2284" and writes "4" to the left of it.

student (T204): Divide?

GPT-5.5: explicit scaffolding and rigor instructions | E18-A | W2 / R2

Exactly—divide, because division undoes multiplication.

So we’re finding:

\(2{,}284 \div 4\)

I set it up as \(4\) into \(2284\). Can you start the long division for me: do we use the first digit \(2\) by itself, or do we need to use \(22\)? Why?

Evidence: do we need to use \(22\)? Why?

Rationale: The learner chooses between two narrowed starting partial dividends and explains the choice using the divisor.

GPT-5.5: general tutoring instructions | E18-B | W1 / R2

Yes, exactly—divide!

Since **4 × ★ = 2,284**, we do the opposite operation:

2,284 ÷ 4 = ★

Let’s do it step by step. **4 can’t go into 2**, so we look at **22**.

How many times does **4** go into **22**?

Evidence: How many times does **4** go into **22**?

Rationale: The initial partial dividend is supplied; the learner calculates its quotient digit.

Human tutor | E18-C | W2 / R2

Yes. We have to divide two thousand two hundred eighty-four by four.

Okay, so what's our first step?

Evidence: what's our first step?

Rationale: The learner determines the appropriate initial step or partial dividend within the already established division setup. That relationship is unsupplied, although the overall strategy is narrowed to division.

E19

Recent conversation

tutor (T355): Integer.

tutor (T356): So integer is any whole number.

tutor (T357): So like fractions are not an integer because it it's not a whole number, right? It's like a part number.

tutor (T358): [PAUSE: 2 seconds]

tutor (T358): So if you double any integer, the answer will be even. So if you times uh any number by two, it'll be will it be even or odd?

tutor (T359): [PAUSE: 6 seconds]

tutor (T359): So yeah, I'll answer that. Uh, if you double any integer, the answer will be even.

student (T360): Never.

tutor (T361): All right.

tutor (T362): [PAUSE: 9 seconds]

student (T362): Huh?

tutor (T363): [PAUSE: 6 seconds]

tutor (T363): Okay. So if you add if you double any integer, it's always even because you're timesing it by two. So let's just watch this.

tutor (T364): [SCREEN_UPDATE] A video plays on the screen. It starts by stating "The statement is always true." It defines an integer as a whole number (positive, negative, or zero) and explains that doubling is multiplying by 2. It gives examples: 2 x 3 = 6 (even), 2 x 4 = 8 (even), 2 x -5 = -10 (even), and 2 x 0 = 0 (even). The video then uses visual counters to show that an odd number (like 3) doubled becomes an even number (6), and an even number (like 4) doubled becomes an even number (8). It concludes by stating the rules: odd x even = even, and even x even = even.

student (T364): Oh.

tutor (T365): [PAUSE: 94 seconds]

tutor (T365): Okay. So, let me just escape. To summarize the video, technically, no matter what number it is, as long as it is a normal whole number, then if you times it by two or double it, because that basically means timesing it by two, it'll always be even.

tutor (T366): So obviously that's a different issue if your number is a fraction or a decimal, because they're not a whole number, right? They're like part of a number. So if you ever multiply fractions or decimals by two, it's not going to have the same rule. It only applies to full numbers. Do you get that?

student (T367): Yeah.

tutor (T368): If you have a question, just ask.

student (T369): Okay.

tutor (T370): [PAUSE: 5 seconds]

tutor (T370): Okay. Which is the largest two-digit even number you think you can make from these tiles that they've given?

student (T371): The largest one that like...

GPT-5.5: explicit scaffolding and rigor instructions | E19-A | W1 / R1

Yeah—the largest two-digit number you can make using the tiles shown, but it still has to be even.

Let’s think it through: for a number to be even, what kind of digit does it need to have in the ones place?

Evidence: what kind of digit does it need to have in the ones place?

Rationale: The learner recalls the even-ones-digit criterion; the task is clear despite the tile values being absent in full preceding context.

GPT-5.5: general tutoring instructions | E19-B | W1 / R2

Yeah—the largest means the biggest two-digit number you can make.

For it to be even, the last digit has to be even: 0, 2, 4, 6, or 8.

So first look at the tiles: which digits are available? Then choose the biggest possible tens digit while still making sure the ones digit is even. What numbers do you see on the tiles?

Evidence: What numbers do you see on the tiles?

Rationale: The rule and maximization strategy are supplied, and the immediate explicit demand is identifying available digits. Missing tile values prevent solving but not classifying that demand.

Human tutor | E19-C | W1 / R2

You said... So if you put these numbers together to make a two-digit number, what can you make that's even? I mean, yeah, that's even.

So you can type in the number that you think.

Okay. Is nineteen an even number?

I mean, like, what do you think?

The best way to check that is, can nineteen be divided by two and still give you a full number?

So nineteen divided by two would be nine point five.

So do you think it's still even?

Evidence: nineteen divided by two would be nine point five

Rationale: The response ultimately shifts from constructing the largest available even number to judging 19. Given its nonwhole quotient and the stated test, only a heavily cued parity recognition remains.

E21

Recent conversation

tutor (T22): [SCREEN_UPDATE] A loading screen for the an online learning platform platform appears, displaying "Angles - Part 2".

tutor (T22): [PROBLEM_CHANGE: p1 - 'Which of the following is true?' (start)] A problem appears on the screen titled "Angles - Part 2". It shows a straight horizontal line with a ray extending upwards and to the left from a point on the line. This creates two angles around the point on the line. The larger, reflex angle is labeled 'p'. The smaller, acute angle is labeled 'q'. The question is "Which of the following is true?". The options are: A) p + q = 90°, B) p + q = 100°, C) p + q = 180°, D) p + q = 360°.

tutor (T22): Okay. So which of the following is true? And then you can let me know which one you think it is.

tutor (T23): [PAUSE: 48 seconds]

student (T23): Um, C.

tutor (T24): [PAUSE: 3 seconds] Tutor clicks on option C, then clicks the "Check" button.

The screen displays a message "C is correct".

Student's cursor points to the reflex angle 'p', then to the straight line itself.

tutor (T24): Okay. Let me put C.

tutor (T25): Yeah. Very good. Can you explain that?

student (T26): So, uh, now you would think it would be more than one eighty because of this part of this of the left side of the angle, this angle. Mmm. Because it has like a much bigger one, but if it's much bigger, like a if the circle is more bigger, it would still be it would still be the regular amount of degrees. And, uh, I know that there's only two angles, so I would just know that two angles make one eighty if they're on a line and they're the only two angles that are on the line.

tutor (T27): Yeah, very good.

tutor (T28): [PAUSE: 6 seconds]

tutor (T28): [PROBLEM_CHANGE: p2 - 'What is the size of angle n?' (start)] Tutor clicks the "Next" button. A new problem appears titled "Angles - Part 2". It shows a straight horizontal line with two rays extending upwards from a point on the line, creating three adjacent angles. The leftmost angle is labeled 132°. The middle angle is labeled 'n'. The rightmost angle is labeled 31.2°. The question is "What is the size of angle n?". The options are: A) 13.2°, B) 16.8°, C) 180°, D) 31.2°.

tutor (T28): Okay, this one. What's the size of angle N?

tutor (T29): [PAUSE: 127 seconds]

tutor (T29): Okay, Jayden, what are you thinking?

tutor (T30): [PAUSE: 25 seconds]

student (T30): Uh, A.

tutor (T31): [PAUSE: 9 seconds]

tutor (T31): Okay, let me check that really quick.

tutor (T32): [PAUSE: 2 seconds]

tutor (T32): [SCREEN_INTERACTION: student - selecting an answer] The student's cursor clicks on option A and then clicks the "Check" button.

tutor (T32): [SCREEN_UPDATE] The screen updates to show "A is correct" along with an explanation: "Angles on a straight line about a point add up to 180°. The 30° does not form a part of a calculation because it is not formed from the same point. The angles we have are n + 48 = 180°. Taking 48 away from both sides gives n = 180 - 48 = 132°."

tutor (T32): Okay, yeah, so it was the correct answer. Do you mind explaining how you got to A?

student (T33): Uh, sure.

GPT-5.5: general tutoring instructions | E21-A | W2 / R1

Great, go ahead. You can start with: “Angles on a straight line add up to ___ degrees,” and then tell me what numbers you used to find \(n\).

Evidence: tell me what numbers you used to find \(n\)

Rationale: The learner must reconstruct the angle relationship and calculations using a supplied sentence scaffold. Full context contains inconsistent diagram/feedback values, but the explanation demand is clear and no candidate numerical answer is asserted.

GPT-5.5: explicit scaffolding and rigor instructions | E21-B | W2 / R1

Go ahead—walk me through your thinking.

Try to include:

1. What do the angles on a straight line add up to?

2. Which angle measures did you already know?

3. What did you do to find \(n\)?

Evidence: What did you do to find \(n\)?

Rationale: Three prompts narrow the explanation to the straight-line total, known angles, and their use; the learner still supplies the reasoning despite conflicting numerical context.

Human tutor | E21-C | W0 / R0

Or if you want some help, I could also chime in.

Evidence: if you want some help, I could also chime in

Rationale: This offers help without independently asking for a new mathematical action; the earlier explanation request is not restated.

E22

Recent conversation

tutor (T197): This one's kind of weird and different, right? What Tom did, he used a trick, I think, to do it in his head easy.

tutor (T198): So it looks like instead of doing eighteen times six, what Tom did is he did first twenty times six, 'cause that's so much easier doing your head. Like I...

tutor (T199): I wouldn't be able to do eighteen, I mean, I'd be like, okay, sixty plus, and then what's like eight times six is what? Six, no, forty-eight, and then I'd be like, okay, like a hundred and eight. Like I would, it would be not super, it would take me several steps, right, to do this part in my head.

tutor (T200): But twenty times six, now what's two times six?

student (T201): Two times six is oh, twelve.

tutor (T202): I was gonna say, you know your doubles. You know that one.

tutor (T203): Okay. So, and then with a zero at the end, so, so if two times six is twelve, twenty times six would be a hundred and twenty. No, no barking. No, no barking. No, no barking. No.

tutor (T204): [PAUSE: 27 seconds]

tutor (T204): Sorry. Oh my goodness. Every time we get a package, it's like the end of the world for my dogs. Okay.

tutor (T205): Okay. So right, you were able to do that in your head like so fast, so easy. So like it kind of makes sense that instead of trying to do eighteen times six, you would do twenty times six, right? Which was like, Toby, don't be naughty. Don't be learning bad lessons from the older dogs. You're just a little puppy. Who are you gonna, what are you gonna do anyways? Sorry.

tutor (T206): Um, but obviously twenty times six is like more than eighteen times six, right? Because twenty is bigger than eighteen.

tutor (T207): So if that's a hundred and twenty, then we know our answer is gonna be less than that. So that's why we're subtracting out a certain amount.

tutor (T208): [PAUSE: 2 seconds] Tutor's cursor hovers over option B (which is 2), then moves and clicks on option A (which is 3).

Tutor clicks the "Check" button. The screen shows "Incorrect" and an explanation.

The screen problem has changed. The new problem is "The members of a tennis club use 84 towels every day. How many towels are used in 7 days?".

Tutor starts writing "4" on the whiteboard to the left of the problem.

Tutor erases the red line she just drew.

tutor (T208): Okay. So we know, so twenty is how many more than eighteen?

student (T209): Twenty is two more than eighteen.

tutor (T210): Oh my gosh, you're so fast at that now. I mean like, sorry. Okay. But anyways, yes. Yes, exactly. Sorry. I know. I'm trying to like be cool and not get too excited, but okay.

tutor (T211): Exactly. So twenty is two more than eighteen. So we got to get rid of, um, so that's, right, I almost accidentally giving you the answer. Um, so if we have twenty times six, but we're actually trying to find out eighteen

times six, how much should we take away to get the real answer?

student (T212): Um,

tutor (T213): It's something times six, right? And we know the twenty was two more than eighteen. So two?

tutor (T214): Yes. Exactly. So, um, so if we put a two here, it'd be like two times six, which you already know is twelve, and you would subtract that from the a hundred and twenty you got before and that would give you your your answer of eighteen times six. Yeah, okay, so two is A. So we'll click A.

tutor (T215): I assume I didn't lead you astray.

tutor (T216): Right. So just like we did. So eighteen is two less than twenty. So if we find the value of twenty sixes and then subtract two of those sixes, then we know what eighteen times six is, right? Because that's what multiplication really is. Like you're saying twenty sixes, but we really only wanted eighteen sixes. So we did that in our head because it was easy, but we need to get rid of those two extra sixes to be the correct answer, right?

tutor (T217): Okay. Go ahead and read this one to me. This lesson was intense. I hope you take a nap after this. You deserve it.

student (T218): The members of a tennis club use eighty-four towels every day. How many towels are used in seven days?

GPT-5.5: explicit scaffolding and rigor instructions | E22-A | W2 / R1

Great reading. Let’s translate the story first.

If they use **84 towels each day** for **7 days**, how many groups of 84 towels is that?

Can you write the multiplication sentence we need to solve?

Evidence: Can you write the multiplication sentence we need to solve?

Rationale: The learner maps daily usage and days to equal groups and a multiplication equation; the equation is not provided.

GPT-5.5: general tutoring instructions | E22-B | W1 / R2

Great. So we have **84 towels each day** for **7 days**.

That means we’re doing **84 seven times**, so what multiplication problem should we write?

Evidence: we’re doing **84 seven times**

Rationale: The decisive repeated-group structure is supplied; translating it into multiplication notation is heavily cued.

Human tutor | E22-C | W0 / R0

Okay. Eighty-four towels, like not per person, I assume. But like, altogether. All right. So how might we...

Go away, you red line.

Evidence: So how might we...

Rationale: The mathematical question trails off before specifying any action, followed by a screen-management remark. The 84-towels-total interpretation agrees with the story.

E23

Recent conversation

student (T213): Nine.

tutor (T214): Good, it's called nine. So if we call this number, Mila, are you with me?

student (T215): Yes.

tutor (T216): It's called 599, okay?

student (T217): Okay.

tutor (T218): Good. Um, so let's try, it's your turn now. I want you to tell me what this number is called and the digits, like the places.

student (T219): I can't see nothing.

tutor (T220): Well, I just wrote it here. Do you see one?

student (T221): Um, um.

tutor (T222): Here, I'll drag it for you. Uh, let's like this one, like this. Does it help?

student (T223): Yes. We need to add up.

tutor (T224): Uh, okay. Oh, this oh, okay. This is cool. Okay. This one. I'll write it big so be as much as you could be.

tutor (T225): Do as much as you can, okay?

student (T226): Okay.

student (T227): It's a 500 and nine tens and nine.

tutor (T228): Now ninety, ninety. Because we have nine tens, it's called ninety.

student (T229): Okay.

tutor (T230): The last one is nine, so it's called 599.

tutor (T231): And what about this one? Uh-uh, no, no, you just erased eight.

tutor (T232): Okay, this one, this number.

student (T233): Um, eight.

tutor (T234): Yes.

tutor (T235): So, can you work with me like this? Like how I explained it to you? Like, oh, this number is uh-huh like verbally, this number is called this, so and so, like that?

student (T236): Yes.

GPT-5.5: general tutoring instructions | E23-A | W1 / R0

Awesome, thank you! Let’s do **801** one step at a time.

Look at the **8** in **801**. Is the 8 in the **hundreds place**, **tens place**, or **ones place**?

Evidence: Is the 8 in the **hundreds place**, **tens place**, or **ones place**?

Rationale: The learner identifies the place of 8 in 801 among named options. Full context confirms 801 was already the displayed current number.

GPT-5.5: explicit scaffolding and rigor instructions | E23-B | W0 / R3

Great. Let’s use the same pattern for **801**.

I’m going to write the place names above it:

Hundreds | Tens | Ones

8 0 1

Start with the **8**: Is the 8 in the **hundreds place**, **tens place**, or **ones place**?

Evidence: **Hundreds | Tens | Ones**

8 0 1

Rationale: The aligned labels explicitly place 8 under Hundreds, supplying the answer before asking for that same place.

Human tutor | E23-C | W0 / R0

Okay.

Evidence: Okay.

Rationale: This acknowledges willingness without eliciting number reading or place-value work.

E25

Recent conversation

tutor (T275): And there's a That's crazy.

student (T276): deer. Well, it's it's the name called

student (T277): Survive ninety nine nights in a in a forest, but there there's a deer that wants to um kill you.

tutor (T278): That's crazy. Did you die?

student (T279): No. I survived until day three. And then I just left the game because she left me.

student (T280): I was scared because I nearly got eaten by the deer because my campsite went out and and I nearly got eaten. So I left the game quick.

tutor (T281): Yeah.

tutor (T282): Yeah, no, that's uh that's messed up to leave you. She shouldn't have done that.

student (T283): It's one of my best friends.

tutor (T284): Oh.

tutor (T285): It is a little bit funny. It's a little bit.

tutor (T286): [PAUSE: 1 second]

tutor (T286): Maybe you'll return the favor one day.

student (T287): Oh trust me, I already left her in one horror game. Pretty much.

tutor (T288): There we go.

tutor (T289): [PAUSE: 2 seconds]

tutor (T289): Okay, can you read this problem for me?

tutor (T290): [PROBLEM_CHANGE: p2 - 'Jack is skip counting in hundreds. 538, 438, [BOX], 238. What number should go in the box?' (start)] The problem is inferred from the dialogue.

student (T290): Jack. Jackson. Michael Jackson.

tutor (T291): Jack.

student (T292): Jack is skip counting in five hun- hundreds.

tutor (T293): There you go.

student (T294): What number should go what number should go in the box?

student (T295): So we're yeah, we're going back down. So it's gonna be B.

GPT-5.5: explicit scaffolding and rigor instructions | E25-A | W1 / R0

Good catch that it’s going back down. What number does choice B show? And what are we subtracting each time?

Evidence: what are we subtracting each time?

Rationale: The learner identifies choice B and the decrement directly specified by counting backward in hundreds; no justification is requested.

GPT-5.5: general tutoring instructions | E25-B | W0 / R3

Exactly—if B is 338, that’s right! We’re going down by 100 each time:

538 → 438 → 338 → 238

Nice job noticing it was going backward. Go ahead and pick B.

Evidence: 538 → 438 → 338 → 238

Rationale: The sequence and missing number are fully supplied; the instruction only selects B. The B mapping is conditional because the packet does not list its value.

Human tutor | E25-C | W0 / R0

Yeah, easy.

Just like the other one, just with a hundred. So, we're doing nice.

Oh, I love that.

That's really cool.

I like his face.

Evidence: Yeah, easy.

Rationale: This confirms B and comments socially without asking for work. The decrement of 100 is valid, but B's value is absent from the available answer-choice description.

E26

Recent conversation

tutor (T154): [PAUSE: 22 seconds]

tutor (T154): So you gotta give me an order. What's the biggest one?

student (T155): C.

tutor (T156): Okay. So that's the biggest. Then what? What's the next biggest?

tutor (T157): [PAUSE: 3 seconds]

student (T157): A.

tutor (T158): Good. Then what?

tutor (T159): [PAUSE: 2 seconds]

student (T159): D.

tutor (T160): And then last?

student (T161): B.

tutor (T162): Good.

tutor (T163): All right, let's watch videos. Let's watch the video on it.

tutor (T164): [PAUSE: 57 seconds]

tutor (T164): [PROBLEM_CHANGE: p1 - 'Who, if either is correct?' (start)] The screen shows "Practice Question 1. Who, if either is correct? Raul and Penny were asked to find the missing number in the calculation ★ x 11 = -165. Raul says the answer is 15 and Penny says it is -15."

tutor (T164): Great. We got it.

tutor (T165): [PAUSE: 6 seconds]

tutor (T165): All right, Kayla, it says, calculate the perimeter of this shape. And remember, each one is three centimeters. So what's the perimeter? Tell me.

tutor (T166): [PAUSE: 6 seconds] Student types "2 cm" in the answer box.

tutor (T166): The perimeter of this whole thing is only two centimeters?

tutor (T167): [PAUSE: 25 seconds] Student deletes "2 cm" from the answer box and types "4".

Tutor writes the numbers 1, 2, 3, 4, and 5 on the screen.

Tutor writes the numbers 6, 7, 8, 9, 10, 11, and 12 on the screen.

tutor (T167): Kayla, let's count. Count with me, okay? One.

tutor (T168): Count.

student (T169): Two, three, four, five.

Human tutor | E26-A | W1 / R0

Keep counting.

Evidence: Keep counting.

Rationale: The learner continues enumerating outside unit edges in the already established perimeter procedure; no new perimeter result is supplied.

GPT-5.5: explicit scaffolding and rigor instructions | E26-B | W2 / R1

Good—keep going around the outside of the shape until you get back to where we started. Don’t stop at 5.

After you count how many little outside sides there are, then ask yourself: if each little side is 3 centimeters, what do we do with the number of sides and 3?

Evidence: what do we do with the number of sides and 3?

Rationale: Besides counting outside edges, the learner must connect edge count and unit length to the perimeter operation.

GPT-5.5: general tutoring instructions | E26-C | W1 / R2

Good—keep going around the outside: six, seven, eight, nine, ten, eleven, twelve.

So we counted **12** outside pieces. Each piece is **3 cm**.

What is **12 × 3**?

Evidence: What is **12 × 3**?

Rationale: The learner computes the specified product. Full context records numbering through 12 but never clearly describes the shape, so the asserted outside-edge count cannot be independently checked.

E27

Recent conversation

tutor (T232): I know you haven't learned division yet, but usually when a number is, you can divide it by two with a, with no remainder. So when you split it in half, each side has equal parts, right? Then it's an even number. So like this one, yeah. So like when you split six into three and three, you see that each side has three blocks each, right?

student (T233): Yeah.

tutor (T234): Yeah. Nice.

tutor (T235): [PROBLEM_CHANGE: p2 - 'Here are 8 blocks. How can the blocks be arranged to show that 8 is even?' (start)] The problem shows 8 blue blocks in a row. Answer choices are A (8 blocks in 2 columns of 4), B (8 blocks in a single row), C (8 blocks in 2 rows of 4), and D (A and C).

Student draws two green arrows on the screen, pointing to split the 8 blocks into two groups of four.

Student clicks on answer choice D.

student (T235): Here are eight blocks. How can the blocks be arranged to show that eight is even?

student (T236): It could be arranged like this. Be arranged. Like this. Four and four.

tutor (T237): Yeah.

tutor (T238): So which of our answers, A, B, or C, or D show that?

student (T239): It's um, C.

tutor (T240): Mhm, C. What about A?

student (T241): It could be A too, so it could be D, A and C.

tutor (T242): Yeah. But what about B?

student (T243): B, it's too many. Too many. One, two, three, four, five, six, seven, eight.

tutor (T244): Hmm.

tutor (T245): Yeah.

tutor (T246): Ready to check?

student (T247): Yeah.

tutor (T248): [SCREEN_INTERACTION: tutor - checks answer] Tutor clicks the "Check" button. The answer D is marked correct.

tutor (T248): Nice.

tutor (T249): How do you feel? Did that make sense?

student (T250): Yeah.

tutor (T251): Yeah.

tutor (T252): [PROBLEM_CHANGE: p3 - '6 is even. [box] + [box] = 6. Which number goes in the boxes to show this?' (start)] The problem shows an equation with two empty boxes adding up to 6. There are 6 red dots to the right. Answer choices are A (3), B (2), C (4), and D (6).

Student draws a green arrow pointing to the first box.

Student clicks on answer choice A.

student (T252): Okay, so six is even.

Human tutor | E27-A | W0 / R0

Mhm.

Evidence: Mhm.

Rationale: This confirms that six is even without eliciting an equation or explanation.

GPT-5.5: explicit scaffolding and rigor instructions | E27-B | W2 / R1

Nice—before we check, tell me your reasoning: if the same number goes in both boxes, what equation would it make, and how does that show 6 is even?

Evidence: what equation would it make, and how does that show 6 is even?

Rationale: The learner must express equal addends and connect that representation to evenness; the explanation remains unsupplied.

GPT-5.5: general tutoring instructions | E27-C | W0 / R3

Exactly — 6 is even because we can split it into two equal groups.

So the two boxes have to be the same number:

3 + 3 = 6

So A, 3, is correct. Nice job! Go ahead and check it.

Evidence: 3 + 3 = 6

Rationale: The equation, evenness reason, and choice are supplied; checking the selected answer introduces no new mathematical action.

E28

Recent conversation

tutor (T43): [PAUSE: 6 seconds]

tutor (T43): Wouldn't it go in? Oh.

tutor (T44): [PAUSE: 30 seconds] The screen shows a fraction strip tool. A 1/6 fraction piece is in the top row of a four-row container.

tutor (T44): Did that work or is that a little too big?

student (T45): Um, I'm not sure.

student (T46): Am I supposed to-

tutor (T47): what happens when you when you put in, like, up on top put in a half and a half and let's see if it gives you some kind of message or if it just... No, I guess if it fits, it fits. Okay.

tutor (T48): [PAUSE: 19 seconds]

tutor (T48): Okay, good.

tutor (T49): [PAUSE: 14 seconds]

tutor (T49): Clear the canvas. Yes. Okay. See if you can get another combination.

tutor (T50): [PAUSE: 5 seconds]

tutor (T50): Okay, good.

tutor (T51): [PAUSE: 7 seconds]

tutor (T51): Super.

tutor (T52): [PAUSE: 9 seconds]

tutor (T52): Okay. Now, suppose I take one of those out. Besides one-fourth, what could you put in there to fill it up?

tutor (T53): Oh, good for you. Okay.

tutor (T54): How about thirds?

tutor (T55): [PAUSE: 10 seconds]

tutor (T55): Oops.

tutor (T56): [PAUSE: 16 seconds]

tutor (T56): what two fractions would you put together to make another third?

student (T57): Um.

GPT-5.5: explicit scaffolding and rigor instructions | E28-A | W1 / R2

No rush. Think of the whole strip cut into **3 equal parts** for thirds.

Now if we take **one of those thirds** and cut *that piece* into **2 equal pieces**, how many equal pieces would the **whole strip** be cut into?

So: 3 parts, each split in half → ___ parts total?

Evidence: 3 parts, each split in half → ___ parts total?

Rationale: The final prompt specifies counting or doubling three equal parts, leaving six to calculate; the conversion to sixths is strongly scaffolded.

Human tutor | E28-B | NA / R0

Yeah, there you go. So, I don't know, I don't know why they don't tell you when it's right or wrong, but that's that's good. No problem there. You got that worked out.

Okay. Okay. How do we stop the browser? I always forget. Okay. Let's, um,

this is just a little practice reducing fractions. So see if you can work those out.

You can write on the whiteboard if you want, or do you have a paper there you're writing on or just thinking about it?

Evidence: this is just a little practice reducing fractions

Rationale: The response switches to reduction exercises that are not included in the preceding context, so their specific demand level cannot be determined. Its success claim also lacks an identified completed strip arrangement.

GPT-5.5: general tutoring instructions | E28-C | W0 / R3

No worries—try using **two 1/6 pieces**.

Because:

1/6 + 1/6 = 2/6, and **2/6 is the same as 1/3**.

Go ahead and drag **two sixths** into the space and see if they match the size of one third.

Evidence: **1/6 + 1/6 = 2/6**

Rationale: The exact pieces, sum, and equivalence to one third are supplied. Dragging those named pieces enacts the given answer without an unsupplied mathematical decision.

E29

Recent conversation

tutor (T154): All right. Go back to our dashboard. Um, I think we speed through the decimal question last time, so hopefully this one will be equally easy, so we can speed through it again as well.

tutor (T155): [PROBLEM_CHANGE: p2 - 'Comparing decimals' (start)] A new problem appears. "What symbol goes in the circle to compare the two decimals? You can use the grids to help you decide." The problem shows two grids, one for 0.35 and one for 0.4, with a blank circle between them. The options are <, >, =, and None.

tutor (T155): Okay, yeah, this one's pretty easy, I think. It is the first question. Looks pretty easy. I like the color blue on the background. I don't think I've ever seen this before.

tutor (T156): Anyway, so for this question, what symbol goes in the circle to compare the two decimal? You can use the grid to help you decide. So the first number is zero point three five. The second number is zero point four. And in the, in the, uh, circle here, you could either put A, which is less than, B, greater than, C, equal, or D, none of them above. So what do you think for this question? I think this one's also pretty straightforward, so I think you got this.

tutor (T157): [PAUSE: 56 seconds]

tutor (T157): So, what do you think?

tutor (T158): [PAUSE: 24 seconds] Tutor's cursor moves over option B and draws a box around it.

tutor (T158): B? Greater than?

tutor (T159): You think point three five is greater than point four?

tutor (T160): [PAUSE: 4 seconds] Tutor's cursor moves from option B to option A.

Tutor draws a box around option A.

Tutor draws a rectangle around the colored portion of the 0.35 grid and another rectangle around the colored portion of the 0.4 grid.

Tutor writes "0.35" and "0.40" on the whiteboard to the right of the problem. Tutor points to the digits while explaining.

Tutor clicks the "Next" button.

tutor (T160): Huh?

tutor (T161): A? You think it's uh less than point four?

student (T162): Okay.

tutor (T163): Let's see if you're right. And you are.

tutor (T164): Point three five is smaller than point four. You can either see it from the grid here. So the grid here is like this. They they nicely colored it for you, so it's a little easier. And point four is right here. You can see point four is bigger than point three.

tutor (T165): Another way to compare them is just to write them side by on top of each other. That's one way to do it as well. Zero point four, zero. Put another zero here as well, just in case. So you can compare from the left to the right. Zero is the same, so you move on to the second number. The second number is a three and a five. And um, since three is smaller than five, that means that point three five is smaller than point four. Good job.

tutor (T166): Next.

tutor (T167): [PROBLEM_CHANGE: p2 - 'Comparing decimals' (start)] A new problem appears. It shows two grids, Grid 1 and Grid 2. Grid 1 has 80 out of 100 squares colored purple. Grid 2 has 59 out of 100 squares colored red. Grid 2 is drawn slightly larger than Grid 1. There are two characters with speech bubbles. Tom says: "0.8 is larger than 0.59, so I did more coloring." Katie says: "Because the grids are different sizes, we can't compare the decimals like that." The question is "Who is correct?" with options: A) Only Tom, B) Only Katie, C) Both Tom and Katie, D) Neither is correct.

Tutor draws an arrow pointing to Grid 1 and another arrow pointing to Grid 2.

tutor (T167): Ooh, okay. We have we have pictures this time. So Tom colored in grid one and Katie colored in grid two. So grid one is right here. This is grid one. Uh grid two is right here. This is grid two. So grid two is slightly bigger than grid one. I don't know why they make this bigger but not the other one bigger. Anyway. So Tom said, point eight is larger than point five nine, so I did more coloring. Uh, Katie said too, because the grids are different sizes, we can't compare the decimals like that. So who is correct? A, only Tom. B, only Katie. C, on both Tom and Katie, or D, neither is correct.

tutor (T168): [PAUSE: 68 seconds]

tutor (T168): What do you think, Maya?

tutor (T169): [PAUSE: 10 seconds] Student clicks on option B and then the "Check" button.

Tutor writes "0.8" and "0.59" on the whiteboard.

tutor (T169): B? Like only Katie is correct? Because the grids are different sizes, we can't compare the decimals like that?

student (T170): Okay, let's see if he's right, she's right.

GPT-5.5: general tutoring instructions | E29-A | W0 / R3

Let’s check it—but I want us to think about it too.

Tom has **0.8**, and Katie has **0.59**.

We can rewrite **0.8** as **0.80**.

So we compare:

0.80 vs **0.59**

That’s **80 hundredths** vs **59 hundredths**.

Since **80 hundredths is more than 59 hundredths**, **0.8 is larger than 0.59**.

So even though the grids look like different sizes on the screen, the decimals tell us the amount: Tom colored more. That means the correct answer should be **A: Only Tom**.

Evidence: the decimals tell us the amount: Tom colored more

Rationale: No work is requested. Although 0.8>0.59 is correct, concluding a larger absolute colored area from unequal-sized wholes is unsupported and selects the wrong claimant.

GPT-5.5: explicit scaffolding and rigor instructions | E29-B | W1 / R2

Before we decide, let’s compare the decimals as numbers.

We can rewrite **0.8** as **0.80**.

So we’re comparing:

0.80 and **0.59**

Which is larger: **80 hundredths** or **59 hundredths**?

Evidence: Which is larger: **80 hundredths** or **59 hundredths**?

Rationale: The learner compares the given integer counts of equal fractional units. The response explicitly restricts itself to numbers and does not assert greater absolute area.

Human tutor | E29-C | W1 / R3

And you're right. Katie has recognized that the grids are different sizes, so we cannot compare the area colored in each sizes.

And yeah, that's one way to we um we need to be mindful of when you're coloring stuff, uh when you're doing grid stuff, because sometimes one grid is slightly bigger and they have different scales, and that can kind of mislead um mislead the the results. So you're going to be careful when you compare them. I think the better comparison is just to write it down. Like Katie is not uh Tom is not wrong that point eight is larger than point five nine, but um because they use different grid sizes, um that kind of makes it a little harder for uh for to compare saying that they he did more coloring. But you don't know which one he did more coloring because grid two is slightly bigger in size. So you don't know which one is referring to what. So that's something to keep mindful of when you when you're doing like coloring grids and stuff, that each grid size must be the same.

Let's move on to the next one. Uh, oh this one's slightly interesting. I don't know what is this. Okay. So it has a water bottle here and some black stuff right there. Okay.

Two calculation are written on a page but water was spilled. Ah, okay, water is spilled. Okay, interesting question. They've been coming up with better question now. The two calculations are written on a page, but water was spilled so some of the numbers are now unreadable. So the question one is point three of whatever this is. Question two is point two five of whatever this is. Which calculation would give the bigger answer? So this is also a very interesting to think about. So do you think it would be A, the question one, B, question two, C, the answer will be equal, or D, we cannot compare as we don't know the numbers. So take a moment, read this question. Uh, it's an interesting question. So take a moment to read about it and let me know what you think the answer is.

You think question one is right? Yeah, not right. You think question one will give a bigger answer?

Let's see if you're right. Uh.

Not quite. I don't believe. Yeah. Not quite. Point three, when you look at it, point three is bigger than point two five, right? So, there's some reasoning behind it that question one would give a bigger answer. But what we need to know is the missing value to finish the calculation. Because we don't know what this missing value is. You get it? So, knowing that,

what do you think the answer would be? Because we don't know what the missing value is. We don't know what this smudge is. And these smudge could be different numbers. So that's a hint that they are, they could be different numbers. So knowing that, what do you think the answer would be?

B, C, or D?

Evidence: these smudge could be different numbers

Rationale: The grid-size reasoning and original answer are supplied, then a different obscured-calculation problem is introduced. The missing-factor issue is strongly explained, leaving recognition of the cannot-compare option.

E30

Recent conversation

tutor (T109): Okay. Okay. Okay.

tutor (T110): Next one. Four times three.

student (T111): Four times three?

tutor (T112): Mhm.

tutor (T113): [PAUSE: 3 seconds]

student (T113): Twelve?

tutor (T114): Yes.

student (T115): Yeah. Eh, eh, eh, eh, eh.

tutor (T116): Okay.

tutor (T117): Are you ready? Next one. Ten times six.

student (T118): Ten times six?

tutor (T119): Yeah.

tutor (T120): [PAUSE: 2 seconds]

student (T120): Sixty?

tutor (T121): Ow. How? Impossible.

student (T122): Did I get it right?

tutor (T123): Yes.

student (T124): Yeah, yeah, yeah, yeah, yeah.

tutor (T125): All right.

tutor (T126): Next one. Seven times three.

student (T127): Seven times three?

tutor (T128): Yes.

tutor (T129): [PAUSE: 2 seconds]

student (T129): Twenty-one?

GPT-5.5: general tutoring instructions | E30-A | W2 / R0

Yes! 7 times 3 is 21. Nice job!

What strategy did you use for that one—did you just remember it, or did you skip-count like 7, 14, 21?

Evidence: What strategy did you use for that one

Rationale: After confirming the known product, the tutor asks the learner to account for their method within remembered-fact or skip-counting possibilities; this is a new explanation demand.

GPT-5.5: explicit scaffolding and rigor instructions | E30-B | W2 / R0

Yes, 21! Nice.

How did you figure out \(7 \times 3\)? Did you count by 7s, count by 3s, or just know the fact?

Evidence: How did you figure out \(7 \times 3\)?

Rationale: The learner must report or explain a method, with three suggested approaches narrowing that account. The confirmed answer is already the learner's.

Human tutor | E30-C | NA / R0

Ah!

You are so smart.

Why? Why? Why, Lucia, are you so smart? Why?

Evidence: Why, Lucia, are you so smart?

Rationale: The why question concerns praise, not the mathematical basis for 7×3; no mathematical demand is established.

E31

Recent conversation

tutor (T27): You sure?

tutor (T28): [PAUSE: 5 seconds] Student enters "8" into the answer box and submits.

student (T28): Eight.

tutor (T29): Eight. Nice job.

tutor (T30): [PROBLEM_CHANGE: p7 - '1 + 4 = ?' (start)] The screen updates to Question 7 of 10: "1 + 4 = ?".

tutor (T30): [PAUSE: 4 seconds] Student enters "5" into the answer box and submits.

student (T30): Five.

tutor (T31): Five. Nice job.

tutor (T32): [PROBLEM_CHANGE: p8 - '5 + 1 = ?' (start)] The screen updates to Question 8 of 10: "5 + 1 = ?".

tutor (T32): [PAUSE: 4 seconds] Student enters "6" into the answer box and submits.

student (T32): Six.

tutor (T33): Six. Nice job.

tutor (T34): [PROBLEM_CHANGE: p9 - '4 + 2 = ?' (start)] The screen updates to Question 9 of 10: "4 + 2 = ?".

tutor (T34): [PAUSE: 3 seconds] Student enters "6" into the answer box and submits.

student (T34): Six.

tutor (T35): Six. Nice job.

tutor (T36): [PROBLEM_CHANGE: p10 - '6 + 4 = ?' (start)] The screen updates to Question 10 of 10: "6 + 4 = ?".

tutor (T36): [PAUSE: 5 seconds] Student enters "10" into the answer box and submits.

Tutor clicks the "KEEP PRACTICING" button, then changes the "NUMBERS COVERED" from "0-10" to "11-20".

Tutor clicks "START THE QUIZ".

student (T36): Ten.

tutor (T37): Ten. Nice job.

tutor (T38): Okay. I don't think we really need to keep going over these 'cause you got all of them right and you did super great.

tutor (T39): Okay, let's go on to the bigger numbers.

tutor (T40): Are you ready?

student (T41): Mhm.

GPT-5.5: general tutoring instructions | E31-A | W0 / R1

Awesome. For these bigger ones, try counting on from the bigger number, or make a 10 if you can. Ready? Let’s do the first one together.

Evidence: Let’s do the first one together.

Rationale: Counting-on and making-ten strategies are offered, but no particular first addition or answer request is supplied. Full preceding context stops before the new quiz's first expression appears.

Human tutor | E31-B | NA / R0

Then let's start in three, two, one, go.

Evidence: three, two, one, go

Rationale: This is a starting countdown without a mathematical prompt; the first larger-number problem is absent from full preceding text.

GPT-5.5: explicit scaffolding and rigor instructions | E31-C | W3 / R0

Awesome. For these bigger ones, I’m going to ask you to explain your strategy sometimes, not just say the answer. Try the first one, and tell me how you got it.

Evidence: Try the first one, and tell me how you got it.

Rationale: The explicit demand is to solve the first larger addition and explain a freely selected strategy. Its operands are not shown, but the open strategy-account demand is clear from the known addition quiz context.

E32

Recent conversation

tutor (T201): Wait, now now... wait, now... let me think. So, we were supposed to go out at seven...

tutor (T202): fifty. So, twenty minutes. Okay. So next next time we'll just we'll just have a slightly longer class to make up for it.

tutor (T203): But I'll let you out a lot earlier this time because I don't want you to miss your party.

student (T204): Okay.

tutor (T205): Okay, wait, then we we do that, we got to cancel this game. We got to start learning immediately.

student (T206): What?

tutor (T207): We got to we got to start learning. I need to shove in as much math as possible.

student (T208): No, you do not need to do it.

tutor (T209): No, we do. We totally do.

student (T210): We don't.

tutor (T211): We do because I get fired if we don't.

student (T212): They're going to be like... So you're so they're going to think that you put like lots of effort into this. Like no fun. No fun at all.

student (T213): If they ask you, Dante, how much have you Mr. Reeves has Mr. Reeves been teaching you, you better say, oh, we learned the entire time. We never ever played games.

student (T214): I could just say that we only played like like learning games.

tutor (T215): Yeah, we only played learning games.

student (T216): We never played any non-learning games.

student (T217): Yeah. Yeah, definitely.

student (T218): Didn't play any learning. Definitely.

tutor (T219): No, no, we definitely, definitely, 100%. Okay. Let's do mental adding and subtracting. So this time we can't use the whiteboard, which is perfect.

student (T220): No.

tutor (T221): So let's start. Uh, what is 10 more than 248?

student (T222): Uh, B.

tutor (T223): Or not B. Not B, not B.

student (T224): Not B. A.

Human tutor | E32-A | W0 / R0

A. Nice.

Evidence: A. Nice.

Rationale: The tutor confirms the selected letter with no work request. Full preceding context explicitly lists A as 258, the correct value of 248+10.

GPT-5.5: general tutoring instructions | E32-B | W1 / R3

Exactly — A, 258. “10 more” means add 10, not add a zero. So 248 goes to 258 because the tens digit goes up by 1.

Quick check: what is 10 more than 361?

Evidence: what is 10 more than 361?

Rationale: The old answer and method are supplied, followed by a distinct specified add-ten calculation.

GPT-5.5: explicit scaffolding and rigor instructions | E32-C | W1 / R2

A is correct — but no more letter guessing. Tell me the number: what is 248 plus 10?

And quick check: when you add 10, which digit should change — the ones, tens, or hundreds?

Evidence: what is 248 plus 10?

Rationale: The learner computes the number rather than only giving A, then recognizes the affected place among named choices. The correct numeric result is not supplied in this response.

E33

Recent conversation

student (T104): Thirty-six.

tutor (T105): So x equals?

student (T106): X equals thirty-six.

tutor (T107): Yes! Good job, Jordan. Good job.

tutor (T108): [SCREEN_UPDATE] Confetti animation plays over the entire screen.

Tutor claps her hands.

student (T108): Yeah.

tutor (T109): Awesome.

student (T110): Okay, there's thirty students and one-fifth is wearing house... hats. How... how many students are not wearing hats?

student (T111): So, thirty minus... no, divided.

tutor (T112): Divided.

student (T113): By one-fifth. And one-fifth by that equals... First, you get six. Six divided by five equals thirty. Then you put down the six.

tutor (T114): That means six of them are wearing hats. How many are not wearing hats?

student (T115): Thirty minus six. Thirty minus six.

tutor (T116): [PAUSE: 11 seconds]

student (T116): It's twenty-seven.

tutor (T117): Twenty-seven? Wait. Thirty minus six is not...

student (T118): Oh! Oh, I got... Oh yeah, that...

tutor (T119): [PAUSE: 8 seconds]

student (T119): Thirty minus six, right?

tutor (T120): Mhm.

student (T121): You can't do that. So take... put down the ten, you get... you get four. Four, put... you get twenty-four.

tutor (T122): [PAUSE: 4 seconds] Tutor points at the student's camera feed.

tutor (T122): Correct.

student (T123): Okay. That's it.

GPT-5.5: general tutoring instructions | E33-A | W0 / R0

Great job finishing all 10, Jordan! Let’s do a quick check before we move on.

For number 5:

$\frac{2}{3} + \frac{1}{3} = \frac{3}{3} = 1$

And I want to fix one thing from earlier: for number 6, $\frac{7}{8}$ is greater than $\frac{3}{4}$, because $\frac{3}{4} = \frac{6}{8}$, and $\frac{7}{8}$ is bigger.

So your answers should be:

1. 26

2. 12

3. 12

4. 7

5. 1

6. 7/8

7. 2/3

8. 32

9. 36

10. 24

Nice work today — especially on the word problems!

Evidence: So your answers should be:

Rationale: All ten answers and two corrections are supplied, with no unsupplied work. Full context verifies the list. Relative to the active hats problem, 24 is confirmation only; the response reopens other earlier tasks.

GPT-5.5: explicit scaffolding and rigor instructions | E33-B | W2 / R2

Nice work finishing all 10, Jordan. Before we call it done, let’s do one quick “prove it” check.

For #10, you got **24** students not wearing hats. Can you explain it in two steps?

1. Why is **one-fifth of 30** equal to **6**?

2. Why do we do **30 − 6** after that?

Evidence: Why do we do **30 − 6** after that?

Rationale: The two operations and their intermediate values are supplied, but the learner must explain fraction-of-a-set and complement reasoning.

Human tutor | E33-C | W0 / R0

Good job, Jordan. Look, Jordan, do you see how easy you're doing all of these?

Evidence: do you see how easy you're doing all of these?

Rationale: This asks for agreement about performance, not a mathematical explanation or new calculation.

E34

Recent conversation

student (T55): Negative two. Negative two is over near the rectangle, I mean triangle.

tutor (T56): Yeah, exactly.

student (T57): Yeah.

tutor (T58): So where would, where would negative one be?

student (T59): Near this part. Right over there. That's where negative one is.

tutor (T60): Yeah. So you have, right, so draw that shape.

student (T61): And that is a shape which is something like... a diamond.

tutor (T62): Yeah.

tutor (T63): [PAUSE: 8 seconds]

tutor (T63): All right. And then let's move on to the next one. Yeah, the next one.

tutor (T64): [PAUSE: 29 seconds] Student erases the diamond they drew, the vertical line on the number line, and the long horizontal line.

tutor (T64): Um, yeah.

tutor (T65): [PAUSE: 18 seconds] Student draws a messy version of the diamond in the box for -1.

tutor (T65): So, any idea what the, what the next one would be?

tutor (T66): [PAUSE: 24 seconds] Student erases the messy diamond and redraws a neater diamond.

Student's cursor points to the square on the number line.

Student attempts to draw the square shape in the box for -1/4.

Student draws a dot inside the shape they drew.

student (T66): Give me a minute. Let me think.

tutor (T67): Okay.

student (T68): So it could, just could, okay? It could be somewhere around... maybe around here, I guess.

tutor (T69): Um, so it's basically somewhere close to...

student (T70): Yeah.

tutor (T71): Yeah, it will be that one. 'Cause negative one over four is very close to zero, right?

student (T72): Yeah, that's what I thought about.

tutor (T73): Yeah, but why would it not be the star?

student (T74): Star is like the other way. It's not the this way. Star is for not like normal. Like, you know.

GPT-5.5: general tutoring instructions | E34-A | W1 / R3

Exactly — the star is on the other side of 0, so it’s positive.

For **negative one-fourth (-1/4)**, we go **one small tick to the left of 0**, and that lands on the **square**. So the square is correct there.

Now let’s do the next blank the same way: find the number, then look straight up to see which shape is on that spot. What number is the next blank asking for?

Evidence: What number is the next blank asking for?

Rationale: The current square answer is supplied; the learner only reads the value of the next blank in the same worksheet. Full context supports a square attempt near zero but does not explicitly verify quarter-tick spacing or the star's exact position.

Human tutor | E34-B | W0 / R2

Yeah, it's for, it's for positive. Yeah.

Evidence: it's for positive

Rationale: The tutor supplies a mathematical interpretation of the star without eliciting work. Its positive position is suggested by the learner but not explicitly fixed in the available diagram description.

GPT-5.5: explicit scaffolding and rigor instructions | E34-C | W2 / R2

Yes — you’re noticing the important idea: **which side of 0** it is on.

Let’s say it more mathematically:

• **Left of 0 = negative**

• **Right of 0 = positive**

So for **-1/4**, start at **0** and move **one quarter step to the left**.

Between **0 and -1**, the spaces are quarters:

0, -1/4, -1/2, -3/4, -1

So look at the first mark to the **left of 0**. What shape is sitting there? And how do you know it’s **-1/4** and not **-1/2**?

Evidence: how do you know it’s **-1/4** and not **-1/2**?

Rationale: The location rule is supplied, but the learner must relate the selected shape and interval position to two fractional values. Full context does not independently specify every tick's scale, so the asserted quarter spacing remains uncertain.

E35

Recent conversation

tutor (T101): When I was in high when I was, well, not in middle not in my middle school, but when I was in high school, we had, I took a cooking class and I took a child development class. So those are like electives that you can pick.

tutor (T102): We even had like baking and pastry. So you would learn how to do that. But that's typically, well, at least that's mostly in high schools.

tutor (T103): Some middle schools might have it, but mine didn't at the time. I think in middle school I took art as an elective. I did art. I did um what is it called? Pottery, like the clay. Ceramics. That's what the class was called.

tutor (T104): I did that in middle school.

tutor (T105): Yeah. It just all depends. You have to see what your school offers and then you take classes based on that.

tutor (T106): And usually they have like languages you can take too. So, that's kind of fun. The older you get, the more you get to choose.

student (T107): I think it's cool that you like get to bake brownies in school and then you just get to eat 'em. Like you can never get hungry between classes. That class you just like, you know, if it's like right before lunch or after.

tutor (T108): I feel different.

student (T109): Exactly.

tutor (T110): No, I love that too. 'Cause my friends would take baking and I took cooking. And I remember one time we did cook in class. I think we made like tortilla chips and like salsa or something. And then I ended up eating it, but there was more time in the class, so she let us bake cupcakes when we were done. So then we got to eat, me and my friends got to make cupcakes at the time after we did cooking.

tutor (T111): 'Cause the same teacher taught cooking and the same teacher taught baking. So that's why.

tutor (T112): But yeah, that was really fun too. And that that's good too, 'cause when you get to middle school or high school, you'll get to choose your own electives and then you'll get to have fun with it. Hopefully, if your school has it.

student (T113): What?

tutor (T114): But yeah. Um, okay. We're going to continue talking about like numerical non-linear sequences. And so let's kind of talk about, refresher on what a linear sequence is.

tutor (T115): [SCREEN_UPDATE] The screen loads a new problem.

tutor (T115): So a linear sequence is a range of numbers and the difference between each term, so the first number, the second number, the third number, um, the difference between each of the numbers is the same. So through addition or subtraction, the numbers are either going up or going down by the same number. And then non-linear would be a sequence of numbers where the the terms are changing but by not the same number through either division or multiplication.

student (T116): Does is that clear? Gotcha.

tutor (T117): Okay.

tutor (T118): [PROBLEM_CHANGE: p1 - '2, 6, ?' (start)] The screen shows a problem: "2, 6, ?". Below it says "A non-linear sequence is found by tripling the previous term. What is the third term of this sequence?". Options are A: 18, B: 9, C: 12, D: 10.

tutor (T118): So our question says, a non-linear sequence is found by tripling the previous term, right? So you multiply the number three times. What is the third, third term of this sequence? So we're saying that the question mark right here is the third term and we're trying to figure out what would it be.

tutor (T119): [SCREEN_INTERACTION: student - selecting answer] Student selects answer A, 18. A checkmark appears next to it.

tutor (T119): Mhm.

tutor (T120): And then how do we know it's eighteen?

student (T121): Um, so this is it's found by tripling the previous term. So then two times three equals six and then six times three equals eighteen.

GPT-5.5: general tutoring instructions | E35-A | W0 / R0

Exactly! You followed the rule perfectly: “tripling” means multiply by 3 each time.

2 × 3 = 6

6 × 3 = 18

So the third term is 18. Nice job! Let’s try the next one.

Evidence: So the third term is 18.

Rationale: This repeats the learner's correct reasoning and result. The next problem is not actually posed.

Human tutor | E35-B | NA / R0

Perfect.

Perfect.

Mhm. The third term can be found by tripling the second term. The second term equals six and three times six equals eighteen. Therefore, the third term must be eighteen.

Our question says, a non-linear sequence is found by halving the previous term. What is the third term of this sequence?

Evidence: found by halving the previous term

Rationale: The tripling explanation is confirmed, then a different halving sequence is introduced. Neither of its starting terms appears in the response or full preceding context, so the specific numerical task cannot be established.

GPT-5.5: explicit scaffolding and rigor instructions | E35-C | W1 / R0

Exactly — nice explanation. You applied the rule twice:

2 → 6 by multiplying by 3, then

6 → 18 by multiplying by 3.

So the third term is 18.

Quick push: if the sequence kept going, what would the 4th term be?

Evidence: what would the 4th term be?

Rationale: The learner continues the same sequence by applying the established multiply-by-three rule to 18. The repeated third-term reasoning adds no new answer information.

E36

Recent conversation

tutor (T253): Oh, those are really cute.

tutor (T254): It's all fruits.

student (T255): Yeah.

tutor (T256): Do they have other kinds?

student (T257): Yeah, they have like, um, avocado. I want the avocado. And there's another one. I don't remember.

tutor (T258): Okay, so what's six times nine?

student (T259): Six times nine equals sixty-eight.

tutor (T260): [PAUSE: 1 second]

tutor (T260): Nope.

student (T261): Six... But why are there so many six? Six times nine.

tutor (T262): Is six the hardest one?

student (T263): Fifty-four. This Gatorade must tell me.

student (T264): I'm I'm going to race quit if this Gatorade doesn't...

tutor (T265): Which of the multiplication... the hardest ones. This Gatorade must tell me.

tutor (T266): Which of the multiplications are the hardest ones for you?

student (T267): I only took a little sip. This this little sip must tell me.

tutor (T268): Mhm.

student (T269): It's equals.

tutor (T270): [PAUSE: 6 seconds] Student writes "54" as the answer to 9 x 6.

student (T270): I know.

student (T271): Fifty-four.

tutor (T272): Mhm.

tutor (T273): [PAUSE: 2 seconds]

student (T273): Can I can I use the calculator on this one right now?

Human tutor | E36-A | W0 / NA

No. The one that we're doing. Just kidding, it's so easy. Sixty.

Evidence: it's so easy. Sixty.

Rationale: The response supplies a result with no action left. Full context ends after 9×6=54; it never identifies any next 'this one' problem. Sixty would be wrong for 9×6, but a possibly newly referenced expression is unavailable.

GPT-5.5: explicit scaffolding and rigor instructions | E36-B | W3 / R1

Not yet — calculator can only be for checking after your brain gives an answer.

Read the problem to me first, and tell me one fact you already know that could help. If it’s close to 6 × 9 = 54, we can build from there.

Evidence: tell me one fact you already know that could help

Rationale: The learner identifies the target and chooses a useful related fact rather than executing an already selected calculation; no target-specific solution is supplied.

GPT-5.5: general tutoring instructions | E36-C | W1 / R1

Not yet—no calculator for the multiplication facts. You can use a strategy though.

Tell me the problem, and we’ll do a quick trick. For example, if it’s a 6 fact, we can use:

6 × 10 = 60, then go down by 6.

Evidence: Tell me the problem

Rationale: The immediate demand is identifying the problem. The tutor offers a ten-times-six-minus-six example but does not ask the learner to choose or execute a target-specific strategy.

E37

Recent conversation

tutor (T103): In other words, thirty divided by three would be ten, right? Divided by three, that'd be ten.

tutor (T104): So we know it wouldn't be, it's gonna be a little bit more.

student (T105): Yeah.

tutor (T106): Do you wanna try B?

tutor (T107): [PAUSE: 3 seconds]

tutor (T107): Thirty-two divided by eight.

tutor (T108): [PAUSE: 4 seconds]

tutor (T108): Okay. And that's because, uh, thirty-two divided by eight is four and four times eight,

tutor (T109): is what? Four times eight is, it's kind of the opposite of it, isn't it?

tutor (T110): [PAUSE: 6 seconds]

tutor (T110): Four times eight is, so if thirty-two divided by eight is four, then four times eight would be, what do you think? Four times eight would be that number, right?

student (T111): Yeah.

tutor (T112): It's kind of like the opposite, it's true, isn't it?

student (T113): Yeah.

tutor (T114): I hope my drawing gets better as the night goes on, Mateo. It's pretty bad tonight. I'm so sorry.

tutor (T115): Okay. Now, okay, let's do this one. I think we're almost done.

tutor (T116): Okay, what is missing from the diagram?

tutor (T117): [PROBLEM_CHANGE: p2 - 'What is missing from the diagram?' (start)] The screen changes to a new problem. The question is "What is missing from the diagram?". A diagram shows a sequence of numbers: 14 -> 21 -> ? -> 35 -> 42. The options are A: 28, B: 24, C: 30, D: 26.

tutor (T117): Oh, it's a pattern. This is a pattern.

student (T118): So fourteen, 'cause

tutor (T119): [PAUSE: 7 seconds]

tutor (T119): What are we adding? Oopsies.

tutor (T120): What are we adding to fourteen to get to twenty-one?

student (T121): 'Cause fourteen plus fourteen equals twenty-eight, but that's after twenty-one.

Human tutor | E37-A | W1 / R0

Oh, that's, that's really smart, Mateo. Uh.

Okay, so fourteen plus fourteen is twenty-eight. So what's twenty-one plus twenty-one?

Evidence: what's twenty-one plus twenty-one?

Rationale: The tutor confirms the learner's 14+14 equality, then asks a separate doubling calculation rather than continuing the +7 missing-term reasoning.

GPT-5.5: general tutoring instructions | E37-B | W0 / R3

Good thinking—28 is involved! Let’s look at how much we add each time:

14 → 21 is +7

21 → ? should also be +7

So 21 + 7 = 28.

And it fits: 28 → 35 is +7, then 35 → 42 is +7. So the missing number is **28**, choice **A**.

Evidence: the missing number is **28**, choice **A**

Rationale: The increment, missing term, subsequent checks, and choice are all supplied without a remaining request.

GPT-5.5: explicit scaffolding and rigor instructions | E37-C | W1 / R2

You're noticing the right number might be 28, but let's prove it from the pattern.

Look at the step from **14 to 21**: how much did it go up by? Try counting: 14, 15, 16… up to 21. How many more is that?

Evidence: Try counting: 14, 15, 16… up to 21

Rationale: The learner computes the step size by a specified count-on procedure; a possible missing number is suggested but its pattern basis still requires this calculation.

E38

Recent conversation

tutor (T344): [PAUSE: 2 seconds]

tutor (T344): Good.

tutor (T345): [PAUSE: 76 seconds]

student (T345): It's

tutor (T346): [PAUSE: 2 seconds]

tutor (T346): Why would that one give the same solution?

student (T347): I miscalculated.

tutor (T348): Okay.

tutor (T349): [PAUSE: 11 seconds]

tutor (T349): If you're stuck, just let me know and I can try to lead you.

tutor (T350): [PAUSE: 15 seconds]

student (T350): I I don't write it. I just like, like get my finger and I just do like, I just I just do like imaginary numbers on my on my table.

tutor (T351): Okay.

tutor (T352): But this one specifically, like you don't need to solve it out. You don't need to solve it out to find the answer to see which ones match.

student (T353): But

tutor (T354): [PAUSE: 2 seconds]

tutor (T354): So it's A.

student (T355): Yes.

tutor (T356): Why?

student (T357): Because the same answer as this is the same answer as this.

tutor (T358): But why is it the same answer? What what do they do from here to here that gives you the same answer?

tutor (T359): [PAUSE: 2 seconds]

tutor (T359): What changes about the ten thousand from here?

student (T360): It's minus one.

Human tutor | E38-A | W1 / R0

Minus one. And then what about this one?

Evidence: And then what about this one?

Rationale: The next operand's change must be determined, following the already narrowed comparison of number changes. The exact other operand is absent even in the available full context, limiting confidence.

GPT-5.5: general tutoring instructions | E38-B | W0 / R3

Exactly — it went from 10,000 to 9,999, so it's minus 1.

Now check the other number too: if that also goes down by 1, then the difference stays the same.

So it's like the two numbers moved together, and the "distance" between them didn't change. That's why A matches.

Evidence: if that also goes down by 1, then the difference stays the same

Rationale: The tutor supplies the equal-shift explanation and says that is why A matches, leaving no requested reasoning. Equal subtraction shifts preserve difference, but full context never shows the actual operation or other operand to verify applicability.

GPT-5.5: explicit scaffolding and rigor instructions | E38-C | W2 / R2

Exactly — 10,000 became 9,999, so it went down by 1.

Now check the other number in A: did it also change by 1? And did it go the right direction so the answer stays the same? Can you say the full reason out loud?

Evidence: did it go the right direction so the answer stays the same?

Rationale: The learner must inspect the other change and connect its direction to preservation of the result. The full expression is missing, so its actual transformation cannot be verified.

E39

Recent conversation

tutor (T34): Try your best, even if you think it's wrong.

tutor (T35): [PAUSE: 10 seconds]

tutor (T35): [SCREEN_INTERACTION: student - writing] Student begins to write the answer for problem 5, starting with the number 3.

tutor (T35): [PAUSE: 29 seconds] The student has written "98,603" on the board.

Tutor highlights the word "NOT" in red.

Tutor circles the hexagon shape in green.

Tutor circles the number "1" next to the first problem.

tutor (T35): Okay, are you done?

student (T36): Mhm.

tutor (T37): Okay, let's go through them together.

tutor (T38): So, can you read me the first problem, please?

student (T39): Which of these does... doesn't... does not have four angles?

tutor (T40): Yeah. And what does four angles mean?

student (T41): Four sides.

tutor (T42): Yeah, good job. So this one definitely does not have four sides. So you are correct.

tutor (T43): So this is correct. Okay, let's go to the second one.

tutor (T44): Go ahead and read it for me.

student (T45): Each box con... contains...

tutor (T46): Contains.

student (T47): contains seven chocolates. How many chocolates will be in six box... boxes?

tutor (T48): Boxes. Yeah. Do you know what contains means?

tutor (T49): [PAUSE: 8 seconds] Tutor circles the number "2" next to the second problem.

Tutor points to the "3" in the fraction "1/3".

tutor (T49): It just means each box has seven chocolates. So that each box has seven chocolates inside. Okay?

tutor (T50): So how do you... how did you get forty-two?

student (T51): Um... I times it.

tutor (T52): Yeah. And why would you times it?

student (T53): 'Cause... 'cause there's six boxes and each box has seven chocolates.

Human tutor | E39-A | W0 / R0

Yeah, good job.

Okay, let's go to the next one.

Evidence: Yeah, good job.

Rationale: This confirms the valid equal-groups explanation. Moving to an unspecified next item does not itself state a mathematical demand.

GPT-5.5: explicit scaffolding and rigor instructions | E39-B | W1 / R2

Exactly — that’s the important idea: **6 boxes** means 6 equal groups, and **7 chocolates in each** group.

What multiplication equation matches that story?

Evidence: What multiplication equation matches that story?

Rationale: The response explicitly gives six equal groups of seven, leaving a strongly cued translation into multiplication notation.

GPT-5.5: general tutoring instructions | E39-C | W1 / R3

Exactly! There are 6 groups of 7, so we multiply \(6 \times 7 = 42\). Nice job explaining your thinking.

Let’s check number 3 together. Can you read number 3 for me, and then tell me: how many equal parts is the rectangle split into?

Evidence: how many equal parts is the rectangle split into?

Rationale: The old multiplication is completed, and the learner counts the partitions in problem 3. Full context supplies the four-part rectangle, so the new recognition/counting demand is established.

E40

Recent conversation

tutor (T142): Right.

tutor (T143): So now let me do another one.

tutor (T144): So I do the shapes all like in twos.

tutor (T145): Okay, now count the total number.

tutor (T146): Of the shapes. All the shapes that are on the board. How many shapes are there? Can you count them for me?

student (T147): Uh, sure.

student (T148): Okay, let's see. One, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen. Fourteen.

tutor (T149): Okay. Is that an even number or an odd number?

student (T150): It's a odd. Even.

tutor (T151): Perfect.

tutor (T152): So what does it mean? It means like when we pair two things together or like club them together, club two things together, meaning pair them, it's always an even number, right? That's what we are seeing.

student (T153): Yeah.

tutor (T154): So that's what the first statement tells us. So what it tells us is that you can pair them all up with none leftover. Do you see what it means now?

tutor (T155): [SCREEN_UPDATE] The screen changes back to the problem. The question is now visible: "Which fact will not help one show that there are an even number of apples?". An image of 12 apples is shown. The student's previous incorrect answer 'A' is highlighted, and a "Try Again" button is shown.

Tutor switches back to the whiteboard.

Tutor draws two circles and a third separate circle.

tutor (T155): Let's do some more drawing. And then possibly I can explain more. So let's do this drawing. Can you draw like how so how will you represent odd numbers, right?

tutor (T156): So let's do this.

tutor (T157): So two. So now we are pairing everything, but let's say like how many balls have I drawn that you have not crossed out? How many balls have I drawn that you have not crossed out?

student (T158): Can you count them for me? Five.

tutor (T159): [PAUSE: 8 seconds] Student draws a small curved line on the whiteboard.

student (T159): Five.

tutor (T160): Very good. So do you see like how one ball doesn't have a partner or a friend or a pair?

student (T161): Yeah.

tutor (T162): Right? So is five odd or even?

student (T163): Odd.

Human tutor | E40-A | W0 / R3

Very good. So what did we learn? We learned that for even numbers, you can pair them together with zero things left, right? But for odd numbers, if you pair everything together, there will be always one thing that's left alone.

Evidence: for odd numbers, if you pair everything together, there will be always one thing that's left alone

Rationale: The opening 'what did we learn' question is immediately answered by the complete parity explanation; only assent remains.

GPT-5.5: general tutoring instructions | E40-B | W0 / R3

Exactly — 5 is odd because one ball is left without a partner.

So remember:

• Even = everyone has a partner, no leftovers.

• Odd = one is left over.

Now let’s go back to the apples question. It asks: “Which fact will NOT help show that the apples are even?”

If it says “pair them up with none left over,” that DOES show even.

If it says “split them into two equal piles,” that DOES show even.

If it says “count by 2s and get to 12,” that DOES show even.

So the answer should be the one that does NOT prove even — the one about the apples being in 3 columns/rows. Can you click that one?

Evidence: the one about the apples being in 3 columns/rows. Can you click that one?

Rationale: The response shifts back to the apples proof-choice problem and supplies both the elimination reasons and target choice. Full context confirms the three-columns option, which alone does not guarantee evenness.

GPT-5.5: explicit scaffolding and rigor instructions | E40-C | W1 / R0

Yes — 5 is odd.

Now tell me why: what happens when you try to make pairs with 5? Do all the balls get a partner, or is one left over?

Evidence: Do all the balls get a partner, or is one left over?

Rationale: The learner recognizes the previously observed leftover in a binary prompt. Although introduced as 'why,' the preceding context already explicitly identifies the unpaired ball and the response narrowly cues that fact.